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PPMUDefTopic

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About the PPMU

About the PPMU
Some UltraFLEX instruments implement four-quadrant pin parametric measurement units (PPMUs), which are used to:

- Force voltage while measuring current (for leakage measurements)
- Force current while measuring voltage (for open/shorts tests)

This section includes the following topic:

- PPMU Specifications for UltraFLEX Instruments
- PPMU Software Licensing Requirements

PPMU information includes the following sections:

- PPMU Programming
- PPMU Debug Display
- PPMU Examples

Related Topics

- PPMU (VBT syntax)

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About the PPMU : PPMU Specifications for UltraFLEX Instruments

PPMU Specifications for UltraFLEX Instruments

Each PPMU is implemented differently on each instrument board. For information on the each instrument's PPMU specifications and hardware implementation, see the following topics:

UltraFLEX PPMU Specifications and Hardware Implementation

Instrument	Specification	Hardware Implementation
AWG6G	Pin Parametric Measurement Unit (PPMU)	AWG6G PPMU
BBAC	Pin Parametric Measurement Unit (PPMU)	
DCIO	Pin Parametric Measurement Unit (PPMU)	DCIO Pin Electronics and PPMU DCIO Merged PPMU
GigaDig	Pin Parametric Measurement Unit (PPMU)	GigaDig PPMU
HPM	Pin Parametric Measurement Unit (PPMU)	HPM PPMU
HSD1000	Pin Parametric Measurement Unit (PPMU)	HSD1000 PPMU
SB6G	Pin Parametric Measurement Unit (PPMU)	SB6G PPMU
TurboAC	PPMU specifications are the same as BBAC, except that the PPMU Vclamp Gain Error is 2% of programmed value, maximum.	
UltraPAC80	Pin Parametric Measurement Unit (PPMU)	
UltraPin800	Pin Parametric Measurement Unit (PPMU)	UltraPin800 PPMU
UltraPin1600	Pin Parametric Measurement Unit (PPMU)	UltraPin1600 PPMU
UltraPin4000	Pin Parametric Measurement Unit (PPMU)	UltraPin4000 PPMU
UltraSerial10G	Pin Parametric Measurement Unit (PPMU)	UltraSerial10G PPMU
VHFAC	Pin Parametric Measurement Unit (PPMU)	VHFAC PPMU

PPMU Programming Parameter Values and Ranges

The following table shows the PPMU parameter limits for UltraFLEX instruments with PPMUs.

UltraFLEX Instrument PPMU Parameter Limits

Instrument	Force Current Values	Force Current Ranges	Force Voltage Values	Measure Current Ranges	VHi Clamp Range	VLo Clamp Range
AWG6G	-50mA to 50mA	20A, 200A, 2mA, 50mA	-1V to 6V	20A, 200A, 2mA, 50mA	-1.2V to 6.2V	-1.2V to 6.2V
BBAC	-2mA to 2mA	200A, 2mA	-2V to 7V	200nA, 2A, 20A, 200A, 2mA	-1.57V to 8.18V	-3.18V to 6.57V
DCIO	-64mA to 64mA[1]	8A, 32A, 128A, 512A, 2mA, 8mA, 32mA	-1V to 6V	8A, 32A, 128A, 512A, 2mA, 8mA, 64mA	-1.3V to 6.3V	-1.3V to 6.3V
GigaDig	-50mA to 50mA	20A, 200A, 2mA, 50mA	-1V to 6V	20A, 200A, 2mA, 50mA	-1.2V to 6.2V	-1.2V to 6.2V
HPM	-30mA to 30mA	20A[2], 200A, 2mA, 30mA	-1V to 1.5V (for 200A and 2mA) 0V to 1.5V (for 30mA) -0.8 to 1.5V (for 20A)	20A, 200A, 2mA, 30mA	-0.8V to 1.5V[3]	-0.8V to 1.5V
HSD1000	-50mA to 50mA[4]	20A, 200A, 2mA, 50mA	-1V to 6V	2A, 20A, 200A, 2mA, 50mA[5]	-2V to 6.5V	-2V to 6.5V
SB6G	-50mA to 50mA	200A, 2mA, 50mA	-1V to 3.6V	200A, 2mA, 50mA	-2V to 4.7V	-2V to 4.7V
TurboAC[6]	-2mA to 2mA	200A, 2mA	-2V to 7V	200nA, 2A, 20A,	-1.57V to 8.18V	-3.18V to 6.57V

				200A, 2mA		
UltraPAC80[7]	-50mA to 50mA	5A, 20A, 200A, 2mA, 50mA	-2V to 7V	5A, 20A, 200A, 2mA, 50mA	-1.57V to 8.18V	-3.18V to 6.57V
UltraPin800	-50mA to 50mA	20A, 200A, 2mA, 50mA	-1V to 6V	2A, 20A, 200A, 2mA, 50mA	-2V to 6.5V	-2V to 6.5V
UltraPin1600	-50mA to 50mA	20A, 200A, 2mA, 50mA	-1V to 6V	2A, 20A, 200A, 2mA, 50mA	-1.5V to 6.5V	-1.5V to 6.5V
UltraPin4000	-30mA to 30mA	200A, 2mA, 30mA	-1V to 1.5V	200A, 2mA, 30mA	-0.8V to 1.5V[8]	-0.8V to 1.5V
UltraSerial10G	-30mA to 30mA	200A, 2mA, 30mA	-1V to 1.5V	200A, 2mA, 30mA	-0.8V to 1.5V[9]	-0.8V to 1.5V
VHFAC	-2mA to 2mA	200A, 2mA	-2V to 7V	200nA, 2A, 20A, 200A, 2mA	-2.403V to 11.347V	-4.013V to 9.737V

1 The 64mA force current value is greater than 32mA force current range because with the DCIO merged PPMU feature, the force current capability is the number of merged channels multiplied by 64mA.

2 When using the HPM 20A range, set the number of samples up to a maximum of 100 to ensure that test results are within tester specifications.

3 The HPM VHi and VLo clamps are not programmable.

4 When the force current is greater than 20mA, the range of voltage values that can be measured is -100mV to 4.5V.

5 If the forcing voltage exceeds 5.5V, current measurement is accurate only up to 1mA, 100mA, and 10mA in the 2mA, 200mA, and 20mA current ranges, respectively. In other words, the three measure current ranges, 2mA, 200mA, and 20mA, are accurate only up to half their full scale value when the forcing voltage exceeds 5.5V. Your measurements in these three ranges should not exceed one-half of the range value.

6 PPMU current forcing and measuring specifications only apply when the voltage clamps are set at least 1 V away from the voltage at the PPMU output.

7 PPMU current forcing and measuring specifications only apply when the voltage clamps are set at least 1 V away from the voltage at the PPMU output.

8 The UltraPin4000 VHi and VLo clamps are not programmable.

9 The UltraSerial10G VHi and VLo clamps are not programmable.

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About the PPMU : PPMU Software Licensing Requirements

PPMU Software Licensing Requirements

Operation of the PPMU does not require a software license.

For more information on IG-XL licensing, see IG-XL Licensing in IG-XL Administration help.

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PPMU Examples

PPMU Examples

This section contains two kinds of program examples for the PPMU:

- IG-XL workbook examples show all IG-XL data, VBT code, and patterns (if used) necessary for a complete running program.
- VBT test instance examples describe the steps for creating a simple DC test flow using the minimum required IG-XL setup.

Workbook Examples

- PPMU Continuity Test Programming
- PPMU Pass/Fail Continuity Test Programming
- PPMU Sequential Leakage Test Programming
- PPMU Parallel Leakage Test Programming
- PPMU Pass/Fail Leakage Test Programming

Test Instance Examples

- PPMU Test Instance Programming
- Optional Instance Parameters for DC Tests

Related Topics

- PPMU (VBT Syntax Reference)

- PPMU Programming

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PPMU Examples : PPMU Continuity Test Programming

PPMU Continuity Test Programming

This workbook example shows how to program a PPMU continuity test for two SN74ALS245 bidirectional digital transceiver devices on two sites. This example uses one HSD1000 board in slot 16.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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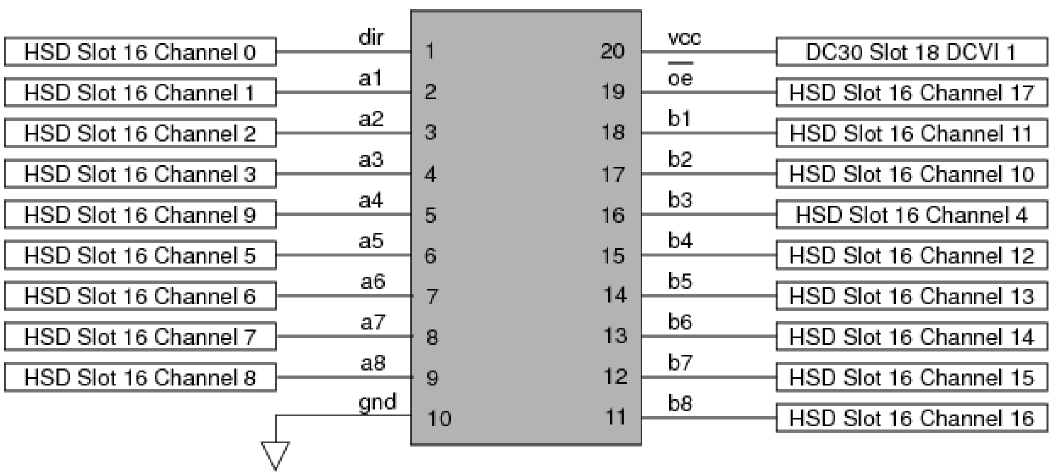
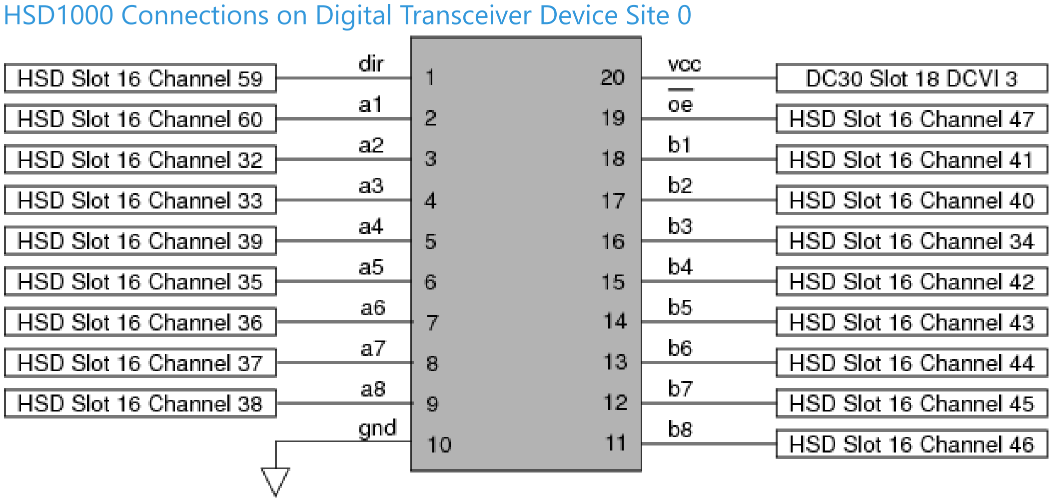
PPMU Examples : PPMU Continuity Test Programming : DUT Connections

DUT Connections



HSD1000 Connections on Digital Transceiver Device Site 0



HSD1000 Connections on Digital Transceiver Device Site 1

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PPMU Examples : PPMU Continuity Test Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	I/O
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc1	Power

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	dir	Input
ctrls	oe	
portA	a1	I/O
portA	a2	
portA	a3	
portA	a4	
portA	a5	
portA	a6	
portA	a7	
portA	a8	
portB	b1	I/O
portB	b2	
portB	b3	
portB	b4	
portB	b5	
portB	b6	
portB	b7	
portB	b8	
ports	portA	I/O
ports	portB	
all_pins	ports	I/O
all_pins	ctrls	

Channel Map Sheet

Channel Map

DIB ID: **View Mode:** Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc1	Pin 21	DCVI	18.sense1	18.sense3

Flow Table Sheet

Flow Table

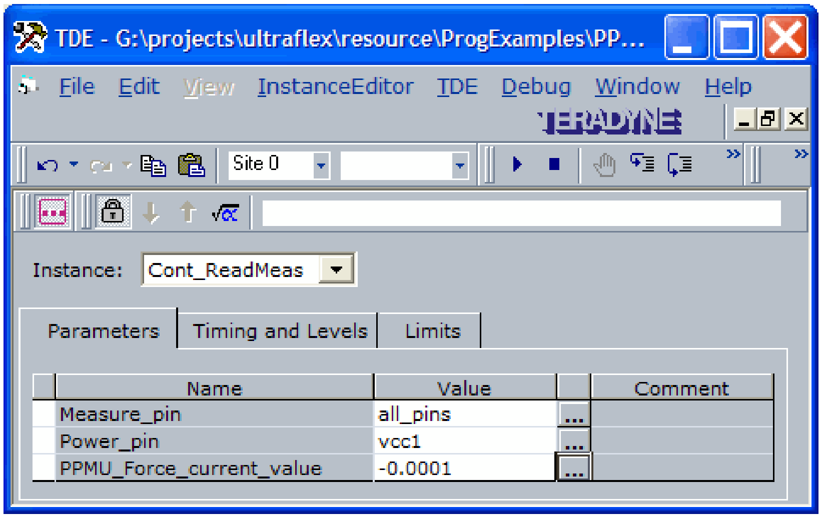
		Gate			Command				Limits		Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	Result
					Test	Cont_ReadMeas	Continuity								
					Use-Limit	Cont_ReadMeas	Continuity		-0.8	-0.2		9		9	Fail
					set-device						1		1		Pass

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
Cont_ReadMeas	VBT	Continuity_Parametric										

The Cont_ReadMeas test instance has the following test-specific parameters and values as shown in the Instance Editor:


TDE - G:\projects\ultraflex\resource\ProgExamples\PP...
File Edit View InstanceEditor TDE Debug Window Help
Site 0
Instance: Cont_ReadMeas
Parameters Timing and Levels Limits
Name Value Comment
Measure_pin all_pins
Power_pin vcc1
PPMU_Force_current_value -0.0001

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PPMU Examples : PPMU Continuity Test Programming : Patterns

Patterns

This PPMU test does not require any pattern files.

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PPMU Examples : PPMU Continuity Test Programming : VBT Code

VBT Code

Option Explicit

```
' Function Name: Continuity_Parametric
' Description: This function tests for opens/shorts (continuity) on the device pins.
' The test connects each device pin to a PPMU and forces a small negative
' current, effectively forward biasing the pin's protection diode to ground.
' The test then simultaneously reads the voltage drop on each pin and
' returns the results for processing.
' Procedure:
' 1) Disconnect device pins from pin electronics.
' 2) Ground the power pin.
' 3) Connect the PPMU to device pins.
' 4) Program PPMU to force current.
' 5) Read and store the voltage results.
' 6) Return results for processing.
' 7) Reset PPMU to default state.
' 8) Return device pins to pin electronics.

Public Function Continuity_Parametric(Measure_pin As pinlist, _
                                      Power_pin As pinlist, _
                                      PPMU_Force_current_value As Double) As Long

    ' Dimension object as PinListData to contain PPMU measured results.
    Dim PPMUMeasure As New PinListData

    On Error GoTo errHandler

    ' Disconnect All_Dig pin electronics from pins to connect PPMUs.
    TheHdw.Digital.Pins(Measure_pin).Disconnect

    ' Set up VCC to 0V.
```

```
    With TheHdw.DCVI.Pins(Power_pin)
        .Gate = False
        .Mode = tlDCVIModeVoltage
        .VoltageRange.Autorange = False
        .VoltageRange.Value = 5 'V
        .CurrentRange.Autorange = False
        .CurrentRange.Value = 0.2 'A
        .Current = 0.2 'A
        .Voltage = 0 'V
        .Connect tlDCVIConnectDefault
        .Gate = True
    End With

   ' Connect the PPMUs to device pins.
   ' Program PPMU Pins to force a small current.
    With TheHdw.PPMU.Pins(Measure_pin)
        .Connect
        .ForceI PPMU_Force_current_value
        .Gate = tlOn
    End With

    ' Make voltage measurements on PPMU pins and store in PinListData PPMUMeasure.
    PPMUMeasure = TheHdw.PPMU.Pins(Measure_pin).Read(tlPPMUReadMeasurements)

    ' Datalog test results.
    TheExec.Flow.TestLimit resultVal:=PPMUMeasure, _
                        unit:=unitVolt, _
                        forceVal:=PPMU_Force_current_value, _
                        forceunit:=unitAmp, Forceresults:=tlForceFlow

    ' Reset PPMUs.
    With TheHdw.PPMU.Pins(Measure_pin)
        .Reset tlResetConnections + tlResetSettings ' Return PPMU connections
                                            ' and settings to default value.
        .Gate = tlOff                          ' For PPMU in HSD1000, default
                                            ' gate value is 'tlOn'.
    End With

    ' Reconnect all measure pins to Pin Electronics.
    TheHdw.Digital.Pins(Measure_pin).Connect

    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
```

End Function

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PPMU Examples : PPMU Pass/Fail Continuity Test Programming

PPMU Pass/Fail Continuity Test Programming

This workbook example shows how to program a PPMU pass/fail continuity test for two SN74ALS245 bidirectional digital transceiver devices on two sites. This test provides a test time improvement since it provides only pass/fail results, not actual values. This example uses one HSD1000 board in slot 16.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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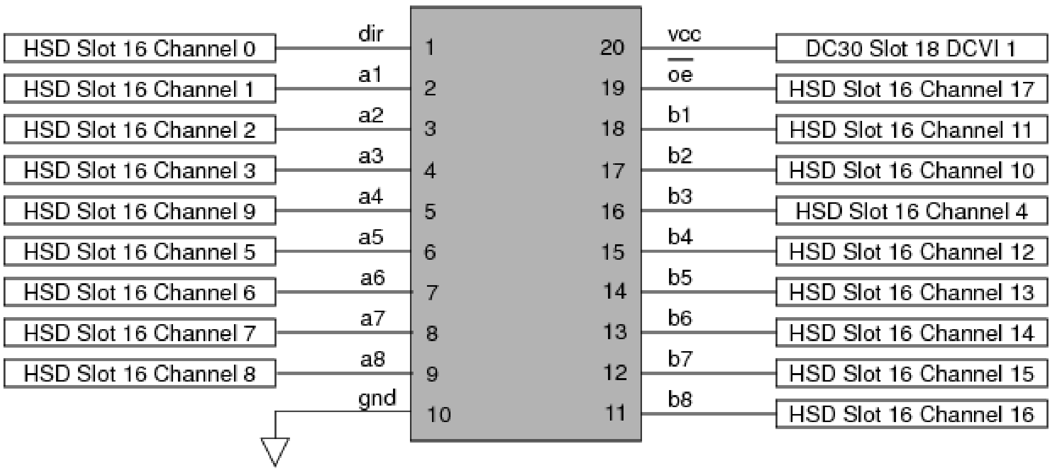
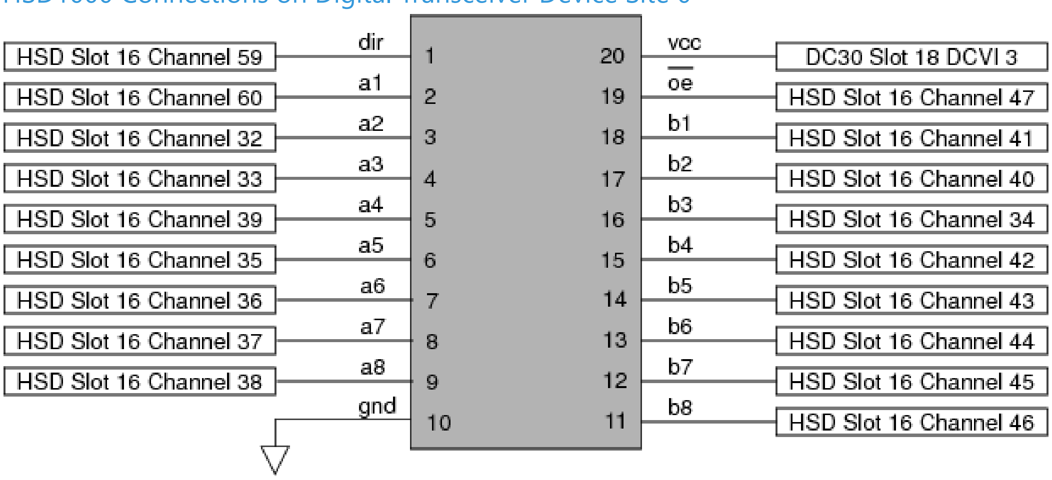
PPMU Examples : PPMU Pass/Fail Continuity Test Programming : DUT Connections

DUT Connections



HSD1000 Connections on Digital Transceiver Device Site 0



HSD1000 Connections on Digital Transceiver Device Site 1

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PPMU Examples : PPMU Pass/Fail Continuity Test Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	I/O
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc1	Power

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	dir	Input
ctrls	oe	
portA	a1	I/O
portA	a2	
portA	a3	
portA	a4	
portA	a5	
portA	a6	
portA	a7	
portA	a8	
portB	b1	I/O
portB	b2	
portB	b3	
portB	b4	
portB	b5	
portB	b6	
portB	b7	
portB	b8	
ports	portA	I/O
ports	portB	
all_pins	ports	I/O
all_pins	ctrls	

Channel Map Sheet

Channel Map

DIB ID: | **View Mode:** Signal

Device Under Test		Tester Channel		
Pin Name	**Package Pin**	**Type**	**Site 0**	**Site 1**
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc1	Pin 21	DCVI	18.sense1	18.sense3

Flow Table Sheet

Flow Table

		Gate			Command				Limits		Datalog Display			Bin		Sort		
Label	**Enable**	**Job**	**Part**	**Env**	**Opcode**	**Parameter**	**TName**	**TNum**	**LoLim**	**HiLim**	**Scale**	**Units**	**Format**	**Pass**	**Fail**	**Pass**	**Fail**	**Result**
					Test	Cont_ReadPassFail	Continuity											
					Use-Limit	Cont_ReadPassFail	Continuity		0	1					9		9	Fail
					set-device									1		1		Pass

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	**Type**	**Name**	**Called As**	**Category**	**Selector**	**Category**	**Selector**	**Time Sets**	**Edge Sets**	**Pin Levels**	**Mixed Signal Timing**	**Overlay**
Cont_ReadPassFail	VBT	Continuity_GoNoGo										

The Cont_ReadPassFail test instance has the following parameters and values as shown in the Instance Editor:

Instance: Cont_ReadPassFail

Parameters | Timing and Levels | Limits

Name	Value		Comment
Measure_pin	all_pins	...	
Power_pin	vcc1	...	
PPMU_Force_current_value	-0.0001	...	
TestLimitVHi	-0.2	...	
TestLimitVLo	-0.8	...	

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PPMU Examples : PPMU Pass/Fail Continuity Test Programming : Patterns

Patterns

This PPMU test does not require any pattern files.

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PPMU Examples : PPMU Pass/Fail Continuity Test Programming : VBT Code

VBT Code

Option Explicit

```
' Function Name : Continuity_GoNoGo
' Description   : This function tests the device pins for opens/shorts (continuity).
' The test connects each device pin to a PPMU and forces a current.
' The test will then read the voltage drop simultaneously on all pins (the
' PPMU determines pass/fail status) and returns the pass/fail result.
' Procedure     : 1) Disconnect device pins from pin electronics.
'                 2) Ground the power pin.
'                 3) Connect the PPMU to device pins.
'                 4) Program PPMU to force a current. The specified upper and lower
'                    limit allow the PPMU to make the pass/fail decision.
'                 5) Read the PPMU pass/fail decision.
'                 6) Return results for processing.
'                 7) Reset PPMU to default state.
'                 8) Return device pins to pin electronics.

Public Function Continuity_GoNoGo(Measure_pin As pinlist, _
                                  Power_pin As pinlist, _
                                  PPMU_Force_current_value As Double, _
                                  TestLimitVHi As Double, _
                                  TestLimitVLo As Double) As Long

    ' Dimension object as PinListData to contain PPMU measured results.
    Dim PPMUMeasure As New PinListData

    On Error GoTo errHandler

    ' Disconnect Measure_pin from pin electronics to connect PPMUs.
    TheHdw.Digital.Pins(Measure_pin).Disconnect
```

```
    ' Set up VCC to 0V.
    With TheHdw.DCVI.Pins(Power_pin)
        .Gate = False
        .Mode = tlDCVIModeVoltage
        .VoltageRange.Autorange = False
        .VoltageRange.Value = 5 'V
        .CurrentRange.Autorange = False
        .CurrentRange.Value = 0.2 'A
        .Current = 0.2 'A
        .Voltage = 0 'V
        .Connect tlDCVIConnectDefault
        .Gate = True
    End With

    ' Connect the PPMUs to Measure_pin.
    ' Program Measure_pin to force current and set TestLimits.
    With TheHdw.PPMU.Pins(Measure_pin)
        .Connect
        .ForceI PPMU_Force_current_value, , TestLimitVHi, TestLimitVLo
        .Gate = tlOn
    End With

    ' Make a Pass/Fail measurement on PPMU pins and
    ' store results in PinListData PPMUMeasure.
    PPMUMeasure = TheHdw.PPMU.Pins(Measure_pin).Read(tlPPMUReadPassFailResults)

    ' Datalog test results.
    TheExec.Flow.TestLimit resultVal:=PPMUMeasure, _
                        forceVal:=PPMU_Force_current_value, _
                        forceunit:=unitAmp, _
                        Forceresults:=tlForceFlow

    ' Reset PPMUs.
    With TheHdw.PPMU.Pins(Measure_pin)
        .Reset tlResetConnections + tlResetSettings ' Return PPMU connections and
                                            ' settings to default value.
        .Gate = tlOff                           ' For PPMU in HSD1000, default
                                            ' value for gate is 'tlOn'.
    End With

    ' Reconnect all measure pins to pin electronics.
    TheHdw.Digital.Pins(Measure_pin).Connect

    Exit Function
  errHandler:
```

```
    If AbortTest Then Exit Function Else Resume Next

End Function
```

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PPMU Examples : PPMU Sequential Leakage Test Programming

PPMU Sequential Leakage Test Programming

This workbook example shows how to program a PPMU sequential leakage test for two SN74ALS245 bidirectional digital transceiver devices on two sites. This example uses one HSD1000 board in slot 16.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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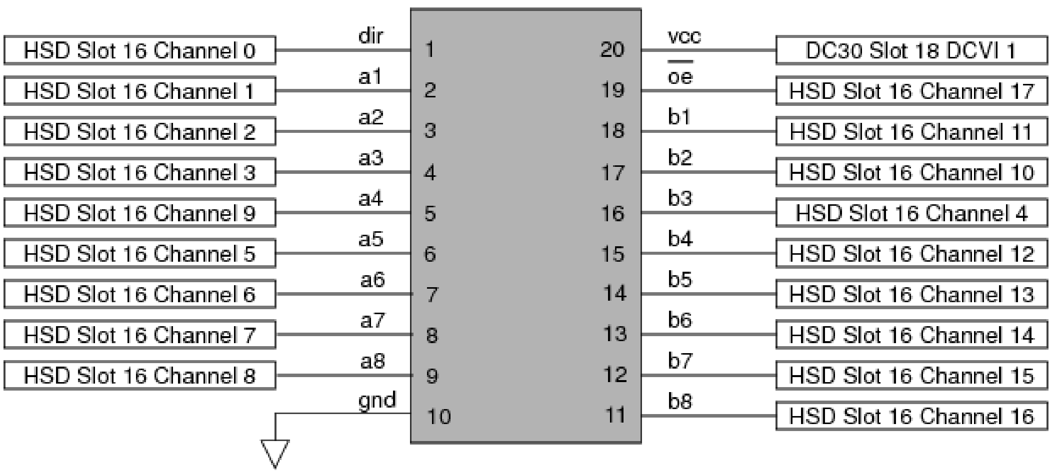
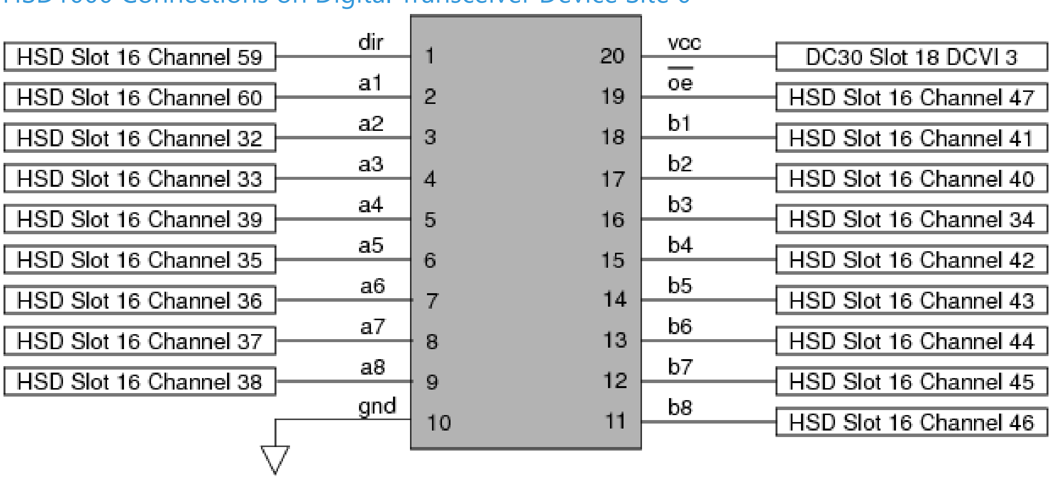
PPMU Examples : PPMU Sequential Leakage Test Programming : DUT Connections

DUT Connections



HSD1000 Connections on Digital Transceiver Device Site 0



HSD1000 Connections on Digital Transceiver Device Site 1

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PPMU Examples : PPMU Sequential Leakage Test Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

•	Pin Map Sheet
•	Channel Map Sheet
•	Flow Table Sheet
•	Pin Levels Sheet
•	Global Specs Sheet
•	Test Instances Sheet

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	I/O
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc1	Power

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	dir	Input
ctrls	oe	
portA	a1	I/O
portA	a2	
portA	a3	
portA	a4	
portA	a5	
portA	a6	
portA	a7	
portA	a8	
portB	b1	I/O
portB	b2	
portB	b3	
portB	b4	
portB	b5	
portB	b6	
portB	b7	
portB	b8	
ports	portA	I/O
ports	portB	
all_pins	ports	I/O
all_pins	ctrls	
odd_portB	b1	I/O
odd_portB	b3	
odd_portB	b5	
odd_portB	b7	
even_portB	b2	I/O
even_portB	b4	
even_portB	b6	
even_portB	b8	

Channel Map Sheet

Channel Map

DIB ID: | **View Mode:** Signal

Device Under Test		Tester Channel		
Pin Name	**Package Pin**	**Type**	**Site 0**	**Site 1**
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc1	Pin 21	DCVI	18.sense1	18.sense3

Flow Table Sheet

Flow Table

TERADYNE

		Gate			Command				Limits		Datalog Display Results			Bin Number		Sort Number		
Label	**Enable**	**Job**	**Part**	**Env**	**Opcode**	**Parameter**	**TName**	**TNum**	**LoLim**	**HiLim**	**Scale**	**Units**	**Format**	**Pass**	**Fail**	**Pass**	**Fail**	**Result**
					Test	Sequential_Leakage	SeqLeak_portB								7		7	Fail
					set-device									1		1		Pass

Pin Levels Sheet

Using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	=200*mA
vcc1		Tdelay	=5*ms
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	=_Vcl_default
all_pins		Vch	=_Vch_default
all_pins		DriverMode	Largeswing-HiZ

Showing evaluated formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	0.2
vcc1		Tdelay	0.005
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	-1
all_pins		Vch	6
all_pins		DriverMode	Largeswing-HiZ

Global Specs Sheet

Global Specs

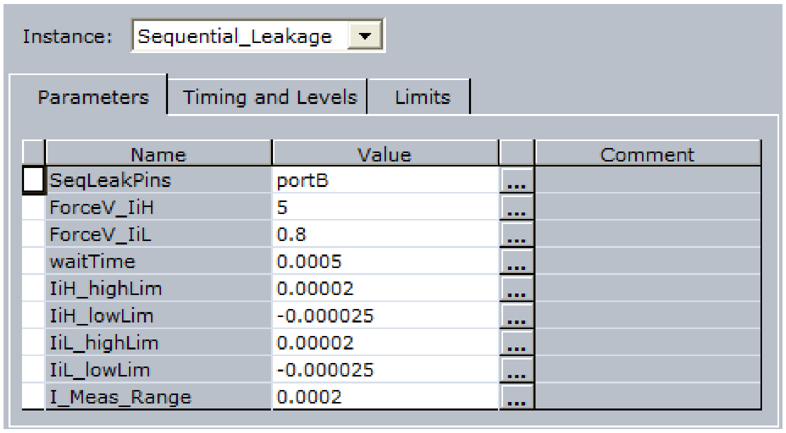
Symbol	Job	Value	Comment
Vcl_default		-1.E+00	
Vch_default		6.E+00	

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
Sequential_Leakage	VBT	SeqLeakage								Levels_Icc		

The Sequential_Leakage test instance has the following test-specific parameters and values as shown in the Instance Editor:


Instance: Sequential_Leakage
Parameters
Timing and Levels
Limits
Name
Value
Comment
SeqLeakPins
portB
ForceV_IiH
5
ForceV_IiL
0.8
waitTime
0.0005
IiH_highLim
0.00002
IiH_lowLim
-0.000025
IiL_highLim
0.00002
IiL_lowLim
-0.000025
I_Meas_Range
0.0002

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PPMU Examples : PPMU Sequential Leakage Test Programming : Patterns

Patterns

This PPMU test does not require any pattern files.

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PPMU Examples : PPMU Sequential Leakage Test Programming : VBT Code

VBT Code

Option Explicit

```
' Function Name: SeqLeakage
' Description  : This function tests for leakage currents on the device input pins.
'               The test is performed sequentially. First, each pin is connected to
'               a logic 1. The PPMU measures the resulting current and returns the
'               reading to the user. Then each pin is connected to a logic 0, the
'               PPMU repeats measurement, and returns the reading to the user.
'  Procedure   : 1) Using pin electronics, precondition the input pins to logic 1.
'               2) Decompose the test pinlist.
'               3) Test for Leakage High on each test pin.
'                  Disconnect pins from pin electronics.
'                  Connect pins to PPMU.
'                  Force pins to logic 1.
'                  Read and store the current measurements.
'                  Return results for processing.
'                  Disconnect PPMUs, and return device pins to pin electronics.
'               4) Test for Leakage Low on each test pin.
'                  Disconnect pins from pin electronics.
'                  Connect pins to PPMU.
'                  Force pins to logic 0.
'                  Read and store the current measurements.
'                  Return results for processing.
'                  Disconnect PPMUs, and return device pins to pin electronics.
'               5) Reset PPMU to default state.
'  Advantages  : Able to measure individual current flow of each pin, and test for
'               pin-to-pin leakage.
' Disadvantages: Time consuming.

Public Function SeqLeakage(SeqLeakPins As Pinlist, ForceV_IiH As Double, _
                           ForceV_IiL As Double, waitTime As Double, _
```

```
                        IiH_highLim As Double, IiH_lowLim As Double, _
                        IiL_highLim As Double, IiL_lowLim As Double, _
                        Optional I_Meas_Range As Double)

    Dim PinArr() As String, PinCount As Long, i As Long

    On Error GoTo errHandler

    ' Connect all signal pins (digital_pins) to the pin electronics and apply levels
    Call TheHdw.Digital.ApplyLevelsTiming(True, True, False, tlPowered, _
      "ports", "ctrls")

    With TheHdw.PPMU.Pins(SeqLeakPins)
      .Gate = tlOff ' Ensure all PPMUs are gated off.
      .Disconnect   ' Disconnect all the PPMUs.
    End With

    ' This Exec function serializes the pins to be tested sequentially.
    TheExec.DataManager.DecomposePinList SeqLeakPins, PinArr(), PinCount

    ' First do Leakage High (ForceV_IiH) test for each pin.
    For i = 0 To PinCount - 1
      Call LeakageSub(PinArr(i), ForceV_IiH, I_Meas_Range, IiH_highLim, _
        IiH_lowLim, waitTime)
    Next i

    ' Next do Leakage Low (ForceV_IiL) test for each pin.
    For i = 0 To PinCount - 1
      Call LeakageSub(PinArr(i), ForceV_IiL, I_Meas_Range, IiL_highLim, _
        IiL_lowLim, waitTime)
    Next i

    ' Reset the PPMU.
    With TheHdw.PPMU.Pins(SeqLeakPins)
      .Reset tlResetConnections + tlResetSettings ' Return PPMU connections and
                                                   ' settings to default state.
      .Gate = tlOff                                ' For PPMU on HSD1000, the
                                                   ' default gate value is 'tlOn'.
    End With

  Exit Function
  errHandler:
    If AbortTest Then Exit Function Else Resume Next

  End Function
```

```
Private Sub LeakageSub(Pin As String, ForceV As Double, I_Meas_Range, _
            HighLim As Double, LowLim As Double, waitTime As Double)

  Dim measVal As New PinListData

  ' Disconnect the pin to be tested from the pin electronics.
  TheHdw.Digital.DisconnectPins (Pin)

  With TheHdw.PPMU(Pin)
    .Connect    ' Connect the PPMU to the DUT pin to be tested.
    .Gate = tlOn ' Gate the PPMU on for the pin to be tested.
    .ForceV ForceV, I_Meas_Range, HighLim, LowLim ' Force voltage, set measure
                        ' range and limits.
    TheHdw.Wait waitTime ' Pause for the force value
            ' to stabilize before measurement.
    measVal = .Read(tlPPMUReadMeasurements) ' Make the measurement.

    ' Test the "measVal" against the limits.
    TheExec.Flow.TestLimit resultVal:=measVal, _
            lowval:=LowLim, _
            hival:=HighLim, _
            unit:=unitAmp, _
            forceVal:=ForceV, _
            forceunit:=unitVolt, _
            Forceresults:=tlForceNone

    .Gate = tlOff ' Gate the PPMU off on the tested pin.
    .Disconnect   ' Disconnect the PPMU from the tested pin.
  End With

  ' Connect the tested pin back to the pin electronics.
  TheHdw.Digital.ConnectPins (Pin)
End Sub
```

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PPMU Examples : PPMU Parallel Leakage Test Programming

PPMU Parallel Leakage Test Programming

This workbook example shows how to program a PPMU parallel leakage test for two SN74ALS245 bidirectional digital transceiver devices on two sites. This example uses one HSD1000 board in slot 16.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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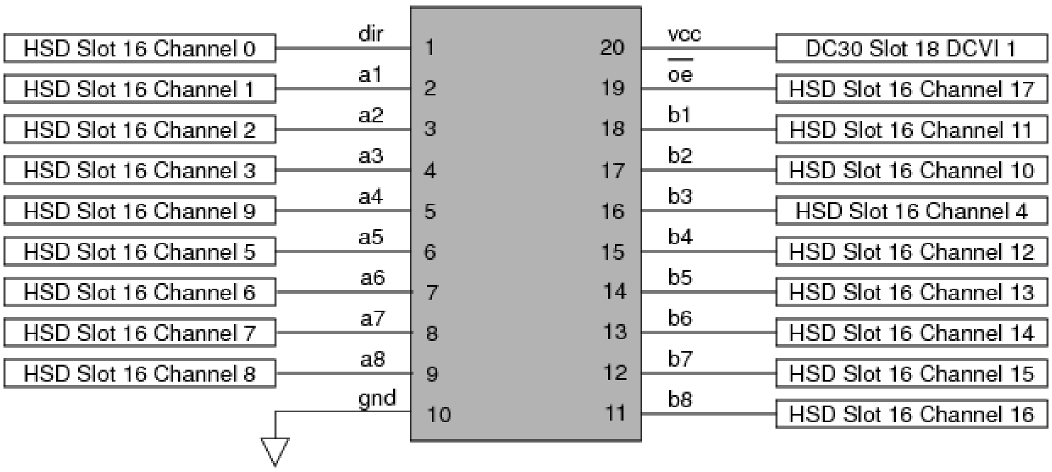
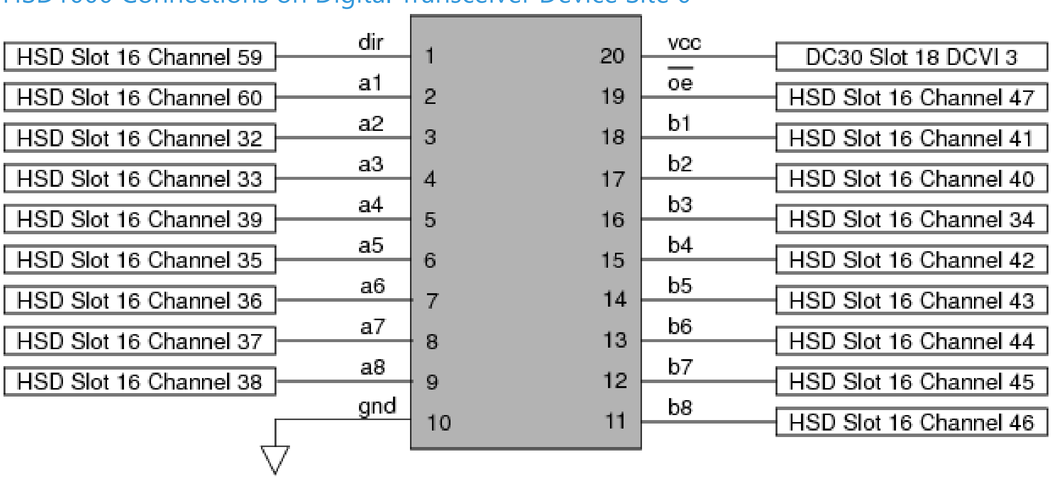
PPMU Examples : PPMU Parallel Leakage Test Programming : DUT Connections

DUT Connections



HSD1000 Connections on Digital Transceiver Device Site 0



HSD1000 Connections on Digital Transceiver Device Site 1

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PPMU Examples : PPMU Parallel Leakage Test Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Pin Levels Sheet
- Global Specs Sheet
- Test Instances Sheet

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	I/O
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc1	Power

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	dir	Input
ctrls	oe	
portA	a1	I/O
portA	a2	
portA	a3	
portA	a4	
portA	a5	
portA	a6	
portA	a7	
portA	a8	
portB	b1	I/O
portB	b2	
portB	b3	
portB	b4	
portB	b5	
portB	b6	
portB	b7	
portB	b8	
ports	portA	I/O
ports	portB	
all_pins	ports	I/O
all_pins	ctrls	
odd_portB	b1	I/O
odd_portB	b3	
odd_portB	b5	
odd_portB	b7	
even_portB	b2	I/O
even_portB	b4	
even_portB	b6	
even_portB	b8	

Channel Map Sheet

Channel Map

DIB ID: **View Mode:** Signal

Device Under Test		Tester Channel		
Pin Name	**Package Pin**	**Type**	**Site 0**	**Site 1**
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc1	Pin 21	DCVI	18.sense1	18.sense3

Flow Table Sheet

Flow Table

		Gate			Command				Limits		Datalog Display Results			Bin Number		Sort Number		
Label	**Enable**	**Job**	**Part**	**Env**	**Opcode**	**Parameter**	**TName**	**TNum**	**LoLim**	**HiLim**	**Scale**	**Units**	**Format**	**Pass**	**Fail**	**Pass**	**Fail**	**Result**
					Test	ParLeak_ReadMeas	Leakage								7		7	Fail
					Use-Limit	ParLeak_ReadMeas	IiH_even_portB		-0.000025	0.00002					7		7	Fail
					Use-Limit	ParLeak_ReadMeas	IiL_odd_portB		-0.000025	0.00002					7		7	Fail
					set-device									1		1		Pass

Pin Levels Sheet

Using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	=200*mA
vcc1		Tdelay	=5*ms
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	=_Vcl_default
all_pins		Vch	=_Vch_default
all_pins		DriverMode	Largeswing-HiZ

Showing evaluated formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	0.2
vcc1		Tdelay	0.005
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	-1
all_pins		Vch	6
all_pins		DriverMode	Largeswing-HiZ

Global Specs Sheet

Global Specs

Symbol	Job	Value	Comment
Vcl_default		-1.E+00	
Vch_default		6.E+00	

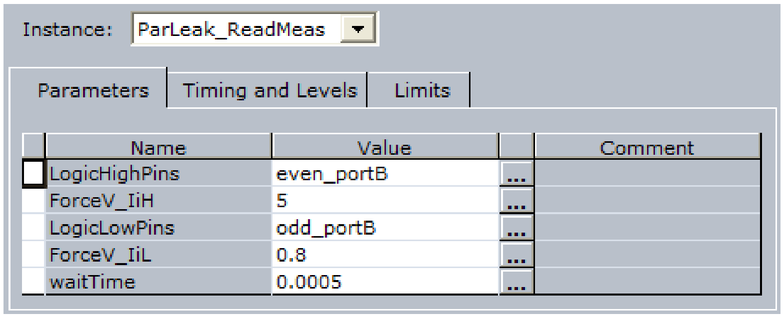
Test Instances Sheet

Test Instances

TERADYNE

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	**Type**	**Name**	**Called As**	**Category**	**Selector**	**Category**	**Selector**	**Time Sets**	**Edge Sets**	**Pin Levels**	**Mixed Signal Timing**	**Overlay**
ParLeak_ReadMeas	VBT	OddEven_Leakage_Parametric								Levels_Icc		

The ParLeak_ReadMeas test instance has the following test-specific parameters and values as shown in the Instance Editor:


Instance: ParLeak_ReadMeas
Parameters
Timing and Levels
Limits
Name
Value
Comment
LogicHighPins
even_portB
ForceV_IiH
5
LogicLowPins
odd_portB
ForceV_IiL
0.8
waitTime
0.0005

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PPMU Examples : PPMU Parallel Leakage Test Programming : Patterns

Patterns

This PPMU test does not require any pattern files.

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PPMU Examples : PPMU Parallel Leakage Test Programming : VBT Code

VBT Code

Option Explicit

```
' Function Name : OddEven_Leakage_Parametric
' Description   : This function tests for leakage currents on device input pins.
' The test is performed by alternately testing even and odd numbered input
' pins, with the PPMU behind each pin forcing either logic 1 or logic 0 level.
' The test returns the resulting measured current on each pin.
' Procedure     : 1) Using pin electronics, precondition the device.
'                 2) Disconnect pins from pin electronics.
'                 3) Connect the PPMU to even numbered input pins.
'                 4) Program PPMU to force even input pins to logic 1.
'                 5) Connect the PPMU to odd numbered input pins.
'                 6) Program PPMU to force odd input pins to logic 0.
'                 7) Read and store the current on even input pins.
'                 8) Read and store the current on odd input pins.
'                 9) Return results for processing.
'                10) Reset PPMU to default state.
'                11) Return device pins to pin electronics

Public Function OddEven_Leakage_Parametric(LogicHighPins As pinlist, _
                                           ForceV_IiH As Double, _
                                           LogicLowPins As pinlist, _
                                           ForceV_IiL As Double, _
                                           waitTime As Double) As Long

    Dim measVal_Hi As New PinListData, measVal_Lo As New PinListData

    On Error GoTo errHandler

    ' Connect all signal pins (digital_pins) to the pin electronics and apply levels
    Call TheHdw.Digital.ApplyLevelsTiming(True, True, False, tlPowered, _
```

```
    "ports", "ctrls")

' Ensure all PPMUs are gated off.
TheHdw.PPMU.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Gate = tlOff

' Disconnect the pin to be tested from the pin electronics.
TheHdw.Digital.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Disconnect

' Force High Voltage.
With TheHdw.PPMU(LogicHighPins)
    .Connect ' Connect the PPMU to the pin to be tested to the DUT.
    .Gate = tlOn ' Gate the PPMU on for the pin to be tested.
    .ForceV ForceV_IiH ' Force voltage.
End With

' Force Low Voltage.
With TheHdw.PPMU(LogicLowPins)
    .Connect ' Connect the PPMU to the pin to be tested to the DUT.
    .Gate = tlOn ' Gate the PPMU on for the pin to be tested.
    .ForceV ForceV_IiL ' Force voltage.
End With

' Pause for the force value to stabilize before measurement.
TheHdw.Wait waitTime

' Perform Leakage measurement.

' Leakage High (ForceV_IiH) Test: Read PPMU value.
measVal_Hi = TheHdw.PPMU(LogicHighPins).Read(tlPPMUReadMeasurements)

' Leakage Low (ForceV_IiL) Test: Read PPMU value.
measVal_Lo = TheHdw.PPMU(LogicLowPins).Read(tlPPMUReadMeasurements)

' Return parametric results to the Flow Table.
TheExec.Flow.TestLimit resultVal:=measVal_Hi, _
                forceVal:=ForceV_IiL, _
                forceunit:=unitVolt, _
                Forceresults:=tlForceFlow
TheExec.Flow.TestLimit resultVal:=measVal_Lo, _
                forceVal:=ForceV_IiH, _
                forceunit:=unitVolt, _
                Forceresults:=tlForceFlow

 ' Reset the PPMU.
With TheHdw.PPMU(LogicHighPins.Value & "," & LogicLowPins.Value)
```

```
        .Reset tlResetConnections + tlResetSettings ' Return PPMU connections and
                                                    ' settings to default state.
        .Gate = tlOff                               ' For PPMU in HSD1000, default
                                                    ' gate value is 'tlOn'.
    End With

    ' Connect the tested pin back to the pin electronics.
    TheHdw.Digital.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Connect

    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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PPMU Examples : PPMU Pass/Fail Leakage Test Programming

PPMU Pass/Fail Leakage Test Programming

This workbook example shows how to program a PPMU pass/fail leakage test for two SN74ALS245 bidirectional digital transceiver devices on two sites. This test provides a test time improvement since it provides only pass/fail results, not actual values. This example uses one HSD1000 board in slot 16.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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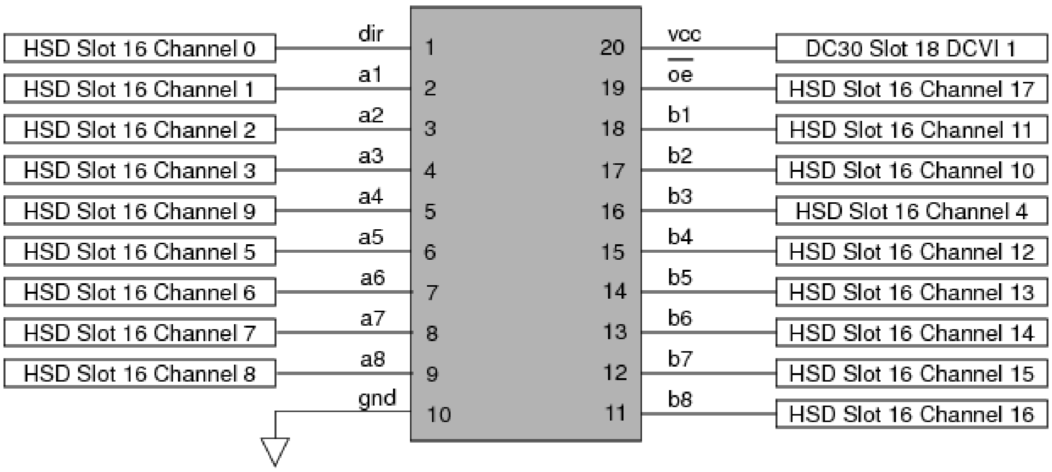
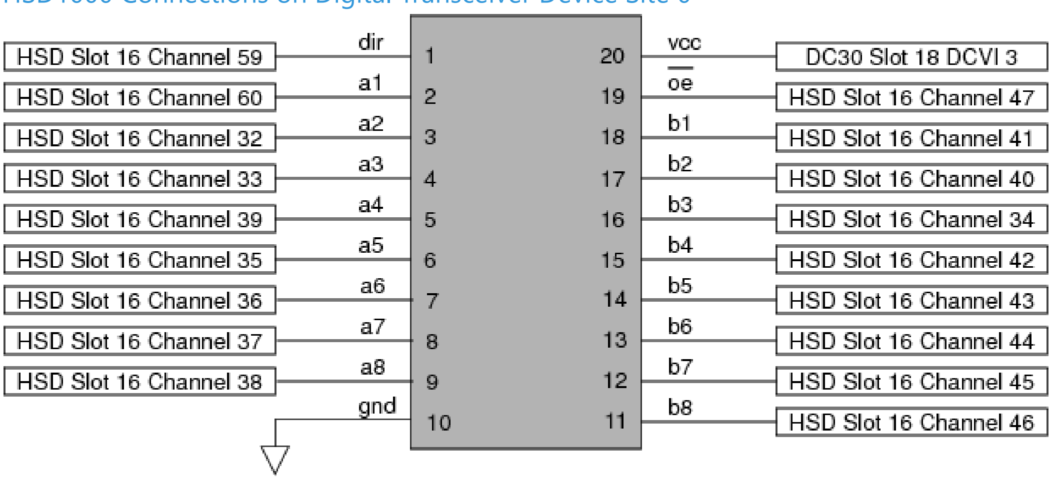
PPMU Examples : PPMU Pass/Fail Leakage Test Programming : DUT Connections

DUT Connections



HSD1000 Connections on Digital Transceiver Device Site 0



HSD1000 Connections on Digital Transceiver Device Site 1

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PPMU Examples : PPMU Pass/Fail Leakage Test Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Pin Levels Sheet
- Global Specs Sheet
- Test Instances Sheet

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	I/O
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc1	Power

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	dir	Input
ctrls	oe	
portA	a1	I/O
portA	a2	
portA	a3	
portA	a4	
portA	a5	
portA	a6	
portA	a7	
portA	a8	
portB	b1	I/O
portB	b2	
portB	b3	
portB	b4	
portB	b5	
portB	b6	
portB	b7	
portB	b8	
ports	portA	I/O
ports	portB	
all_pins	ports	I/O
all_pins	ctrls	
odd_portB	b1	I/O
odd_portB	b3	
odd_portB	b5	
odd_portB	b7	
even_portB	b2	I/O
even_portB	b4	
even_portB	b6	
even_portB	b8	

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	**Package Pin**	**Type**	**Site 0**	**Site 1**
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc1	Pin 21	DCVI	18.sense1	18.sense3

Flow Table Sheet

Flow Table

TERADYNE

		Gate			Command				Limits		Datalog Display Results			Bin Number		Sort Number		
Label	**Enable**	**Job**	**Part**	**Env**	**Opcode**	**Parameter**	**TName**	**TNum**	**LoLim**	**HiLim**	**Scale**	**Units**	**Format**	**Pass**	**Fail**	**Pass**	**Fail**	**Result**
					Test	ParLeak_ReadPassFail	Leakage								7		7	Fail
					Use-Limit	ParLeak_ReadPassFail	liH_even_portB		0	1					7		7	Fail
					Use-Limit	ParLeak_ReadPassFail	liL_odd_portB		0	1					7		7	Fail
					set-device									1		1		Pass

Pin Levels Sheet

Using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	=200*mA
vcc1		Tdelay	=5*ms
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	=_Vcl_default
all_pins		Vch	=_Vch_default
all_pins		DriverMode	Largeswing-HiZ

Showing evaluated formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc1		Vps	5
vcc1		Isc	0.2
vcc1		Tdelay	0.005
all_pins		Vil	0
all_pins		Vih	4
all_pins		Vol	1
all_pins		Voh	2
all_pins		Vt	1.5
all_pins		Vcl	-1
all_pins		Vch	6
all_pins		DriverMode	Largeswing-HiZ

Global Specs Sheet

Global Specs

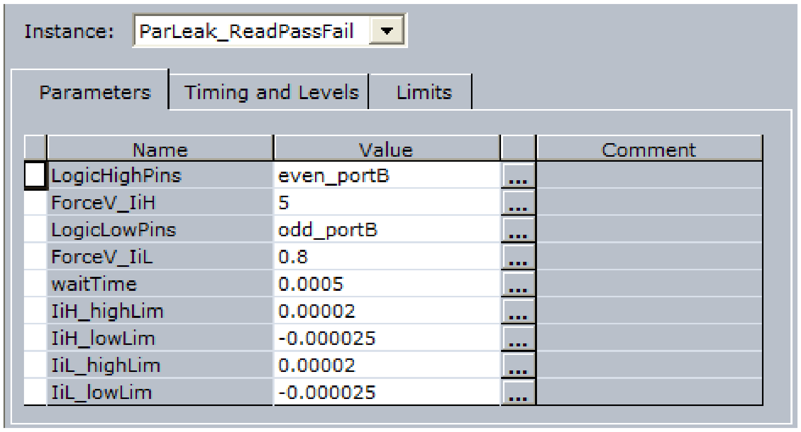
Symbol	Job	Value	Comment
Vcl_default		- 1.E+00	
Vch_default		6.E+00	

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
ParLeak_ReadPassFail	VBT	OddEven_Leakage_GoNoGo								Levels_Icc		

The ParLeak_ReadPassFail test instance has the following test-specific parameters and values as shown in the Instance Editor:



Name	Value		Comment
LogicHighPins	even_portB	...	
ForceV_IiH	5	...	
LogicLowPins	odd_portB	...	
ForceV_IiL	0.8	...	
waitTime	0.0005	...	
IiH_highLim	0.00002	...	
IiH_lowLim	-0.000025	...	
IiL_highLim	0.00002	...	
IiL_lowLim	-0.000025	...	

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PPMU Examples : PPMU Pass/Fail Leakage Test Programming : Patterns

Patterns

This PPMU test does not require any pattern files.

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PPMU Examples : PPMU Pass/Fail Leakage Test Programming : VBT Code

VBT Code

Option Explicit

```
' Function Name : OddEven_Leakage_GoNoGo
' Description   : This function tests for leakage currents on the device input pins.
' The test is performed by alternately testing odd and even numbered input
' pins, with the PPMU behind each pin forcing either logic 1 or logic 0.
' The resulting current is measured. The PPMU determines a pass or fail and
' reports the decision to the user.
' Procedure     : 1) Using pin electronics, precondition the device.
'                 2) Disconnect pins from pin electronics.
'                 3) Connect the PPMU to even numbered input pins.
'                 4) Program PPMU to force even input pins to logic 1.
'                    Since PPMU will decide pass/fail, set current upper limit =20uA
'                    and lower limit =-25uA.
'                 5) Connect the PPMU to odd numbered input pins.
'                 6) Program PPMU to force odd input pins to logic 0.
'                    Since PPMU will make a pass/fail decision, set current upper
'                    limit =20uA and lower limit =-25uA.
'                 7) Read the current on even input pins, PPMU decides pass or fail.
'                 8) Read the current on odd input pins, PPMU decides pass or fail.
'                 9) Return results for processing.
'                 10) Reset PPMU to default state.
'                 11) Return device pins to pin electronics.

Public Function OddEven_Leakage_GoNoGo(LogicHighPins As PinList, _
                        ForceV_IiH As Double, _
                        LogicLowPins As PinList, _
                        ForceV_IiL As Double, _
                        waitTime As Double, _
                        IiH_highLim As Double, _
                        IiH_lowLim As Double, _
```

```
                        IiL_highLim As Double, _
                        IiL_lowLim As Double) As Long

Dim measVal_Hi As New PinListData, measVal_Lo As New PinListData

On Error GoTo errHandler

' Connect all signal pins (digital_pins) to the pin electronics and apply levels
Call TheHdw.Digital.ApplyLevelsTiming(True, True, False, tlPowered, _
    "ports", "ctrls")

' Ensure all PPMUs are gated off.
TheHdw.PPMU.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Gate = tlOff

' Disconnect the pin to be tested from the pin electronics.
TheHdw.Digital.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Disconnect

' Force High Voltage.
With TheHdw.PPMU(LogicHighPins)
    .Connect ' Connect the PPMU to the pin to be tested to the DUT.
    .Gate = tlOn ' Gate the PPMU on for the pin to be tested.
    ' Force voltage, set measure range to AutoRange and limits.
    .ForceV ForceV_IiH, , IiH_highLim, IiH_lowLim
End With

' Force Low Voltage.
With TheHdw.PPMU(LogicLowPins)
    .Connect ' Connect the PPMU to the pin to be tested to the DUT.
    .Gate = tlOn ' Gate the PPMU on for the pin to be tested.
    ' Force voltage, set measure range to AutoRange and limits.
    .ForceV ForceV_IiL, , IiL_highLim, IiL_lowLim
End With

' Pause for the force value to stabilize before measurement.
TheHdw.Wait waitTime

' Perform Leakage Measurement

' Leakage High (ForceV_IiH) Test: Read PPMU. Decide pass or fail.
measVal_Hi = TheHdw.PPMU(LogicHighPins).Read(tlPPMUReadPassFailResults)

' Leakage Low (ForceV_IiL) Test: Read PPMU. Decide pass or fail.
measVal_Lo = TheHdw.PPMU(LogicLowPins).Read(tlPPMUReadPassFailResults)

' Return pass/fail result to the Flow Table.
```

```
    TheExec.Flow.TestLimit resultVal:=measVal_Hi, _
                    forceVal:=ForceV_IiL, _
                    forceunit:=unitVolt, _
                    Forceresults:=tlForceFlow
    TheExec.Flow.TestLimit resultVal:=measVal_Lo, _
                    forceVal:=ForceV_IiH, _
                    forceunit:=unitVolt, _
                    Forceresults:=tlForceFlow

    ' Reset the PPMU.
    With TheHdw.PPMU(LogicHighPins.Value & "," & LogicLowPins.Value)
      .Reset tlResetConnections + tlResetSettings ' Return PPMU connections and
                                          ' settings to default state.
      .Gate = tlOff                           ' For PPMU in HSD1000, default
                                          ' gate value is 'tlOn'.
    End With

    ' Connect the tested pin back to the pin electronics.
    TheHdw.Digital.Pins(LogicHighPins.Value & "," & LogicLowPins.Value).Connect

    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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PPMU Examples : PPMU Test Instance Programming

PPMU Test Instance Programming

This section contains an example test program that demonstrates the use of Pin PMU test. The program contains:

- Four HSD1000 pins
- Four PPMU tests that force current and measure voltage
- Five PPMU tests that force voltage and measure current

Tests in this example test program use all available current ranges on the HSD1000 PPMU. Basic DC testing using the Pin PMU test can be performed with only a few sheets and no VBT code.

To program a simple DC test flow using the PinPmu_T test instance, follow this basic procedure:

1. Define Pins
2. Map Channels to Pins
3. Create Test Instances
4. Add Instances to Flow
5. Set Test Limits

Once you complete this procedure, you can validate and run the test program.

Define Pins

Begin by defining the pins on a Pin Map sheet. This test program includes four pins. Since they are HSD1000 pins, define them as type I/O. A pingroup called ALLPINS contains all four pins.

Pin Map

Group Name	Pin Name	Type
	pin0	I/O
	pin1	I/O
	pin2	I/O
	pin3	I/O
ALLPINS	pin0	I/O
ALLPINS	pin1	
ALLPINS	pin2	
ALLPINS	pin3	

Pin Map Sheet

Map Channels to Pins

Next, create the channel map. The channels used in the test program depend on the configuration of the interface and test system on which test program will be run. In this test program, the pins are assigned to the first four channels on the HSD1000 board in test head slot 10.

Channel Map

DIB ID: View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
pin0		I/O	10.ch0	
pin1		I/O	10.ch1	
pin2		I/O	10.ch2	
pin3		I/O	10.ch3	

Channel Map Sheet

Create Test Instances

Create tests using the Pin PMU test on a Test Instances sheet:

1. Enter the name of the test you want to create on the first empty row of the Test Instances sheet.
2. Select VBT in the Type column and the PinPmu_T under the Name column.

These are the only required entries for a basic DC test. Other optional columns are described later. The example below shows several new test instances of the PinPmu_T test.

Test Instances

	Test Procedure	
Test Name	Type	Name
FIMV_FixRange_50mA	VBT	PinPmu_T
FIMV_FixRange_2mA	VBT	PinPmu_T
FIMV_FixRange_200uA	VBT	PinPmu_T
FIMV_FixRange_20uA	VBT	PinPmu_T
FVMI_FixRange_50mA	VBT	PinPmu_T
FVMI_FixRange_2mA	VBT	PinPmu_T
FVMI_FixRange_200uA	VBT	PinPmu_T
FVMI_FixRange_20uA	VBT	PinPmu_T
FVMI_FixRange_2uA	VBT	PinPmu_T

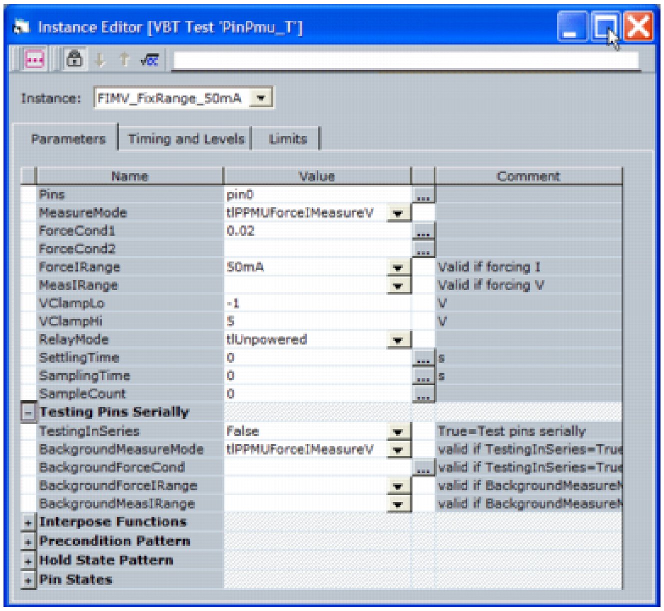
Test Instances Sheet

3. Once you define the test name and type, right-click the instance name and select Edit to open the instance editor.

This opens the instance editor in TDE. The instance editor is where you enter all parameters that describe the test.

4. Enter all PPMU-specific parameters that describe the test in the instance editor.

For information on each parameter field, see Pin PMU VBT Test.

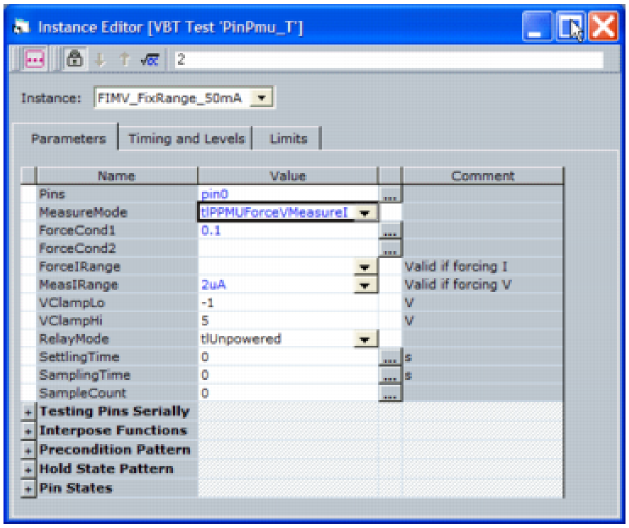


PPMU Instance Editor

You must specify the following parameters to create a DC test (note that some of these parameters are classified as optional because they have defaulted values):

Pins	All the pins to be tested. You can enter a list of multiple pins separated by commas, and you can also use pingroup names. You can type pin names directly, or you can use the ellipsis button (...) to the right of the Pins field to select from the pins available in the test program.
MeasureMode	Select whether this test measures voltage or current.
ForceCond1	Program the force current or force voltage value in the this field. Use either explicit values or equations with spec variables. Click the ellipsis button (...) to access an equation editor.
ForceIRange	Select the force current range. If you select the default value of Auto, the template tries to select the correct range. For best performance, however, explicitly specify a range. The HSD1000 PPMU has four force current ranges: 20A, 200A, 2mA, and 50mA.
MeasIRange	Select the measure current range. If you select the default value of Auto, the template tries to select the correct range. For best performance, however, explicitly specify a range. The HSD1000 PPMU has five measure current ranges: 2A, 20A, 200A, 2mA, 50mA.

The following figure shows an example of an instance that forces 0.1V and measures current in the 2A range.



PPMU Instance Editor: Force Voltage, Measure Current

You program test limits after adding the instances to test flow. See Set Test Limits.

Basic DC testing using the Pin PMU does not require timing or levels. If the test requires running a pattern before or during the test, however, then the levels, timing, and pattern used by the test must be created and included in the test program. Timing and levels are described in Timing and Levels.

Add Instances to Flow

Once you create the test instances, add the tests to a Flow Table sheet. To add a test instance to a row in the flow table:

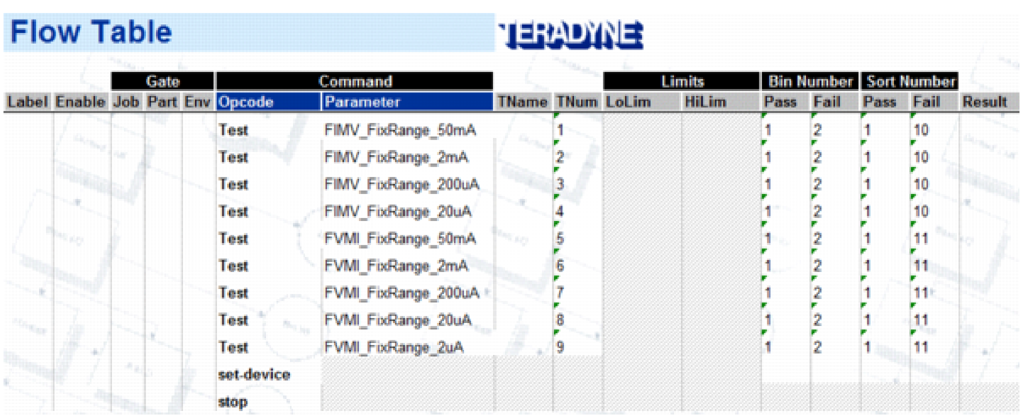
1.	Right-click a cell in the row and select Insert Test.

The Insert Test window appears. This window lists all test instances on the Test Instances sheet.

2.	Select the name of the test instance to add and click OK.

This adds the Test keyword in the Opcode column and the instance name in the Parameter column.

3.	Repeat this procedure for all tests to be run.



Flow Table

		Gate			Command				Limits		Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	Result
					Test	FIMV_FixRange_50mA		1			1	2	1	10	
					Test	FIMV_FixRange_2mA		2			1	2	1	10	
					Test	FIMV_FixRange_200uA		3			1	2	1	10	
					Test	FIMV_FixRange_20uA		4			1	2	1	10	
					Test	FVMI_FixRange_50mA		5			1	2	1	11	
					Test	FVMI_FixRange_2mA		6			1	2	1	11	
					Test	FVMI_FixRange_200uA		7			1	2	1	11	
					Test	FVMI_FixRange_20uA		8			1	2	1	11	
					Test	FVMI_FixRange_2uA		9			1	2	1	11	
					set-device										
					stop										

Flow Table Sheet with Test Instances Added

Optionally, you can define test numbers and test bins and sorts in the Flow Table. In this example test program, nine PPMU test instances are included in the flow. Note that test, bin, and sort numbers are specified.

Set Test Limits

Once you add the test instances to the flow table, you can program the test limits for each instance in the flow. To set the test limits:

1.	Right click the instance name in the Parameter column of the Flow Table sheet and select Edit>Instance.

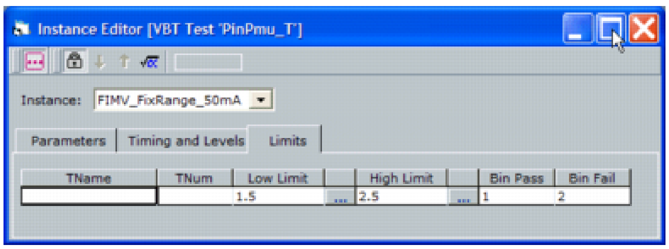
This opens the instance editor in TDE.

☑ **Note:**

You cannot add test limits by editing an instance from the Test Instances sheet. By specifying limits only in the test flow, you can execute the same test instance multiple times using different limits.

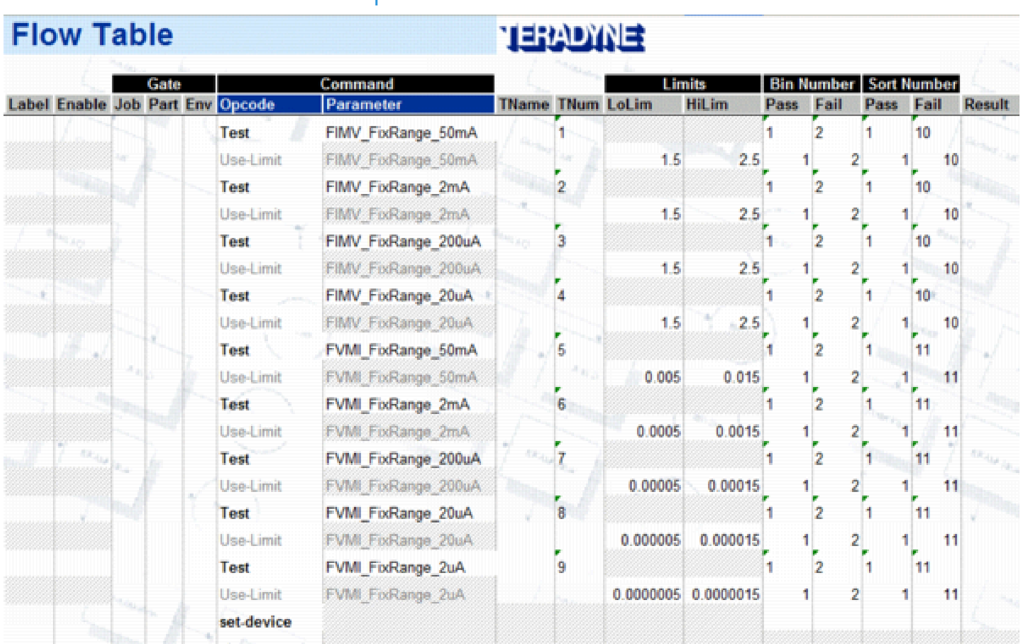
2.	Click the Limits tab and enter the high and low test limits.

You can explicitly program numerical limits or use equations. Selecting the ellipsis button to the right of the limit field opens the equation builder. You can use any spec variables that exist in the test program, and the equation builder lists all available specs.



PPMU Instance Editor, Limits Tab

Once you add test limits to the instance, the Flow Table sheet shows these limits on a row below the test instance name that uses the Use-Limit opcode.



Flow Table

		Gate			Command				Limits		Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	Result
					Test	FIMV_FixRange_50mA		1			1	2	1	10	
					Use-Limit	FIMV_FixRange_50mA			1.5	2.5	1	2	1	10	
					Test	FIMV_FixRange_2mA		2			1	2	1	10	
					Use-Limit	FIMV_FixRange_2mA			1.5	2.5	1	2	1	10	
					Test	FIMV_FixRange_200uA		3			1	2	1	10	
					Use-Limit	FIMV_FixRange_200uA			1.5	2.5	1	2	1	10	
					Test	FIMV_FixRange_20uA		4			1	2	1	10	
					Use-Limit	FIMV_FixRange_20uA			1.5	2.5	1	2	1	10	
					Test	FVMI_FixRange_50mA		5			1	2	1	11	
					Use-Limit	FVMI_FixRange_50mA			0.005	0.015	1	2	1	11	
					Test	FVMI_FixRange_2mA		6			1	2	1	11	
					Use-Limit	FVMI_FixRange_2mA			0.0005	0.0015	1	2	1	11	
					Test	FVMI_FixRange_200uA		7			1	2	1	11	
					Use-Limit	FVMI_FixRange_200uA			0.00005	0.00015	1	2	1	11	
					Test	FVMI_FixRange_20uA		8			1	2	1	11	
					Use-Limit	FVMI_FixRange_20uA			0.000005	0.000015	1	2	1	11	
					Test	FVMI_FixRange_2uA		9			1	2	1	11	
					Use-Limit	FVMI_FixRange_2uA			0.0000005	0.0000015	1	2	1	11	
					set-device										
					stop										

Flow Table Sheet with Defined Test Limits

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PPMU Examples : Optional Instance Parameters for DC Tests

Optional Instance Parameters for DC Tests

You can program several optional PPMU parameters during a DC test, including the following:

- Global Specs
- Voltage Clamps and Settling Time
- Timing and Levels
- Patterns
- Pin Start/Init States
- Interpose Functions

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PPMU Examples : Optional Instance Parameters for DC Tests : Global Specs

Global Specs

You can specify many parameters in a Pin PMU test as an equation using global specs. Each test program workbook can contain one Global Specs sheet. You can use the spec variables defined on this sheet anywhere in the test program. The sheet below shows seven specs available for use in the test program (five default specs and two added for this test program).

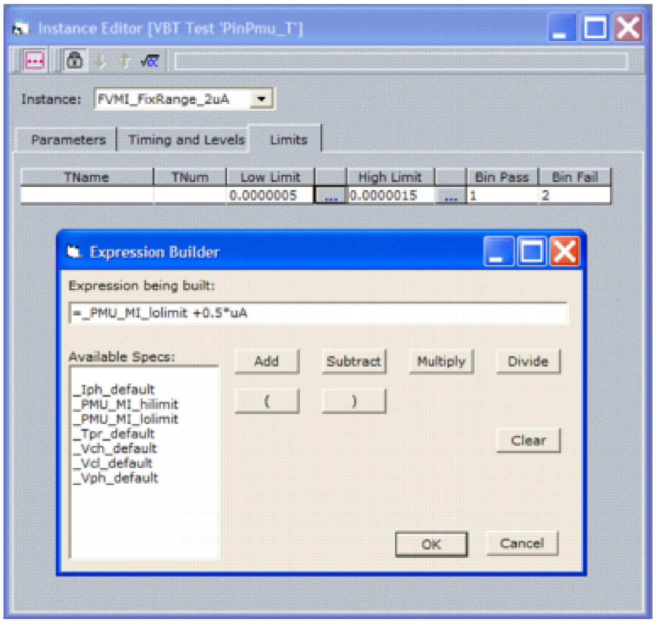
Global Specs

Symbol	Job	Value	Comment
Vcl_default		- 1.E+00	Detector clamp voltage low
Vch_default		5.E+00	Detector clamp voltage high
Vph_default		5.E+00	Hi-V pin voltage high
Iph_default		0	Hi-V pin current limit
Tpr_default		16.E-06	Hi-V pin rise time
PMU_MI_hilimit		1.5E-06	High limit for PMU current measurements
PMU_MI_lolimit		0	Low limit for PMU current measurements

Global Specs Sheet

In this test program, two specs, PMU_MI_hilimit and PMU_MI_lolimit are used to set the test limits. When using spec variables in equations, the spec name must be preceded by an underscore (_) symbol. To define the test limit using these spec variables, click the ellipsis button (...) to the right of the Low Limit or High Limit fields. This brings up the Expression Builder, which lists the available specs to use in the expression (see figure below).

Note that once you have created an expression, the instance editor shows the value of the equation in the parameter fields unless you click the Formula button.



PPMU Instance Editor: Setting Test Limits

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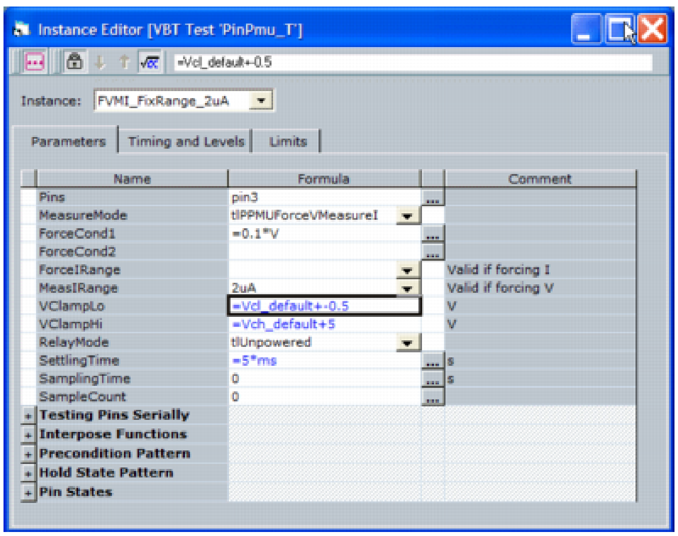
PPMU Examples : Optional Instance Parameters for DC Tests : Voltage Clamps and Settling Time

Voltage Clamps and Settling Time

You can program voltage clamps to ensure that the PPMU does not go outside a specified voltage range. Set the clamp levels in the VClampHi and VClampLo parameter fields. You can explicitly program the values or use global specs.

In addition, you can program settling time to cause the PPMU to pause between forcing a value and making the measurement. In the example below, the PPMU forces 0.1V, waits 5ms, and then makes a current measurement. You can also use global specs in any equation here.



PPMU Instance Editor: Setting Voltage Clamps

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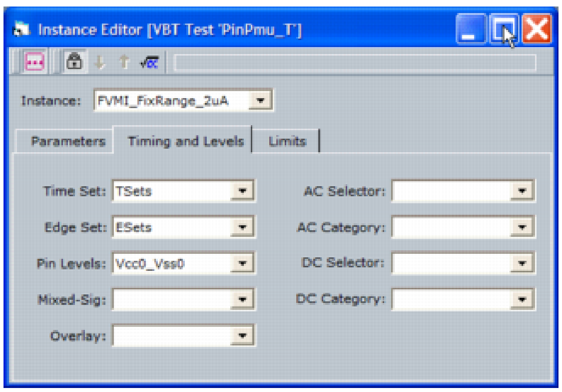
PPMU Examples : Optional Instance Parameters for DC Tests : Timing and Levels

Timing and Levels

You can apply timing and levels during a PPMU test. Typically, this is only required if a pattern is run before or during a PPMU measurement. Levels may be needed if any pins are initialized to either high or low states.

To include desired levels and timing in a PPMU test instance, select the Timing and Levels tab of the instance editor. Drop-down menus allow you to select from a list of available timesets, edgesets, and levels. The example test program contains only one of each, so those have been selected in this instance.



PPMU Instance Editor: Timing and Levels

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PPMU Examples : Optional Instance Parameters for DC Tests : Patterns

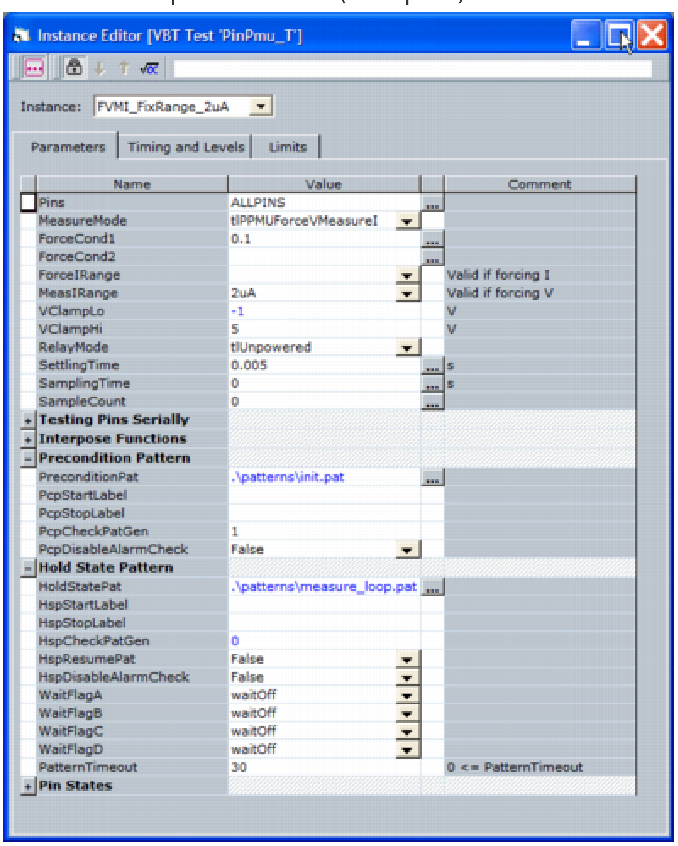
Patterns

You can run a precondition pattern, a hold state pattern, or both during a PPMU test.

- A precondition pattern runs before IG-XL takes the PPMU measurement.

- A hold state pattern runs in a loop while IG-XL makes the measurement.

To specify the pattern to be run, select the Precondition Pattern and Hold State Pattern sections of the instance editor and enter the pattern name (with path) in the field.



PPMU Instance Editor: Specifying Pattern To Be Run

For both types of patterns, you can optionally specify the start and stop labels. If you do not specify labels, the pattern runs from the beginning to end. In addition, if you enable the PcpCheckPatGen or HcpCheckPatGen fields (1 = enable), then the test will fail if the pattern fails. When these parameters are disabled (0 = disable), the instance will fail only if the PPMU measurement fails, regardless of what the pattern result is.

Hold state patterns generally contain a loop that keeps the PatGen running while the PPMU makes the measurement. In the simplest implementation, set all WaitFlag parameters to WaitOff, and the pattern loops while the PPMU makes measurement. After the measurement, IG-XL forces the PatGen to halt. The example in the figure above shows this usage of the hold state pattern.

If you must resume a hold state pattern after making the PMU measurement, you must write the pattern to loop on a WaitFlag condition. When the flag condition is met and the loop begins, IG-XL gives control to the test instance for the PPMU measurement to take place. Once the PPMU makes the measurement, if HspResumePat is set to True, IG-XL changes the WaitFlag condition so that the pattern can resume until the end (or stop vector, if specified).

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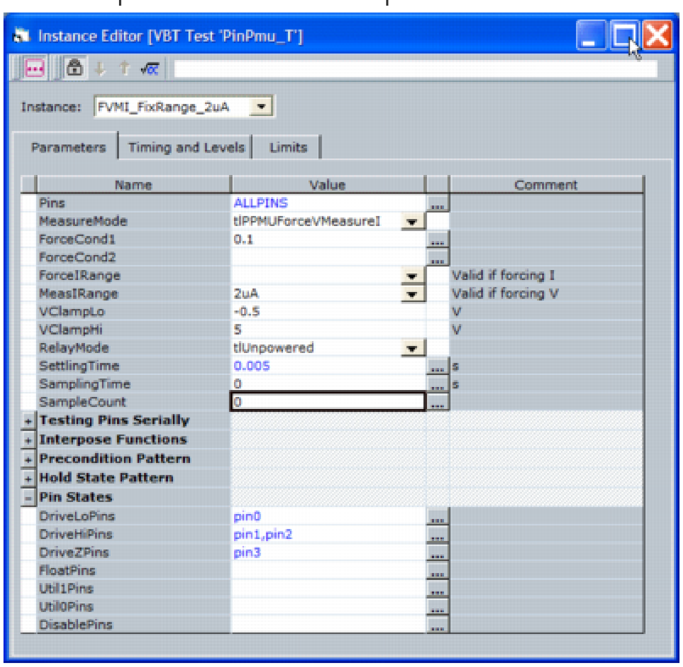
PPMU Examples : Optional Instance Parameters for DC Tests : Pin Start/Init States

Pin Start/Init States

You can set pins to drive either high, low, or off using the DriveHiPins, DriveLoPins, and DriveZPins parameters. Specifying pins in these fields sets the pins’ start state and init state. Start state is the state the pin has at the beginning of a pattern, and the init state is the state that is programmed during the test instance prebody before IG-XL runs any patterns or makes any measurements.

Set pin states by opening the Pins State section of the instance editor. Entering a pin name, a pingroup name, or list of pins and pingroups sets both the start and init states of those pins. You can also float pins by listing them in the FloatPins parameter. Do not list pins to be measured in the FloatPins field.



PPMU Instance Editor: Start and Init States

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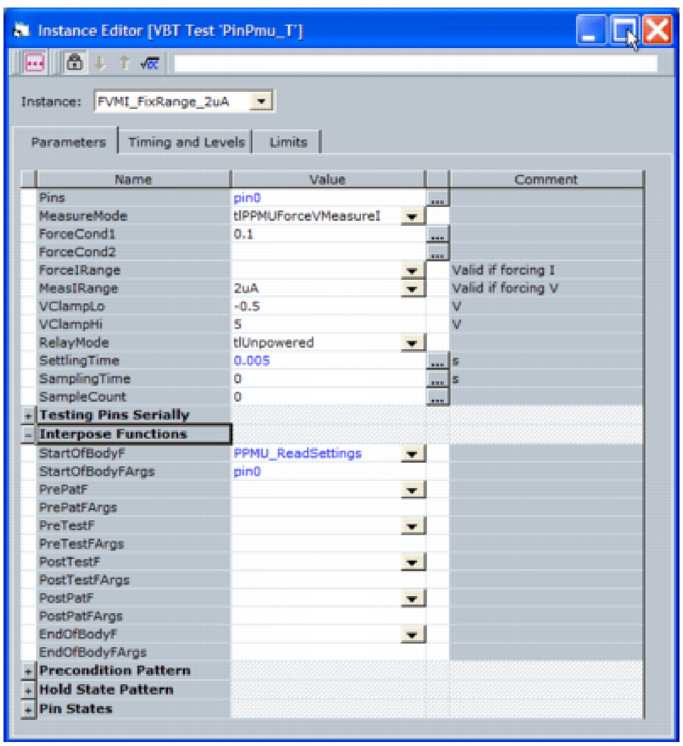
PPMU Examples : Optional Instance Parameters for DC Tests : Interpose Functions

Interpose Functions

The Pin PMU test instance provides several interpose points at which user functions can be executed. Select the Interpose Functions section of the instance editor to see the available interpose points. Entering the name of a VBT function in an interpose function field executes that function at the appropriate point during the test instance execution. You can pass in parameters to the functions by placing them into the <InterposePoint>Args field under each function name.

This example calls the function PPMU_ReadSettings with the pin0 argument passed to it at the StartOfBodyF interpose point. IG-XL calls this function after the prebody is complete, before executing any code in the test instance body.



Instance Editor: Interpose Function

The following is the interpose function called by the instance, as well as the private function that it calls in turn:

Public Function PPMU_ReadSettings(argc As Long, argv() As String)

Dim pinnames As String

pinnames = argv(0)

Call PPMUReadSetting(pinnames)

End Function

```
Private Function PPMUReadSetting(pinname As String)

' ForceV
With TheHdw.PPMU.Pins(pinname)
If .Mode = tlPPMUForceVMeasureI Then
   TheExec.Datalog.WriteComment ("Mode = ForceVMeasureI")
   TheExec.Datalog.WriteComment ("Force Voltage = " & .Voltage)
   TheExec.Datalog.WriteComment ("MeasureCurrentRange = " & .MeasureCurrentRange)
   TheExec.Datalog.WriteComment ("Limit IHi = " & .TestLimits.IHi)
   TheExec.Datalog.WriteComment ("Limit ILo = " & .TestLimits.ILo)
   TheExec.Datalog.WriteComment ("Measured Current = " & .Read(tlPPMUReadMeasurements))
End If

' ForceI
If .Mode = tlPPMUForceIMeasureV Then
   TheExec.Datalog.WriteComment ("Mode = PPMUForceIMeasureV")
   TheExec.Datalog.WriteComment ("Force Current = " & .Current)
   TheExec.Datalog.WriteComment ("ForceCurrentRange = " & .forceCurrentRange)
   TheExec.Datalog.WriteComment ("Limit VHi = " & .TestLimits.VHi)
   TheExec.Datalog.WriteComment ("Limit VLo = " & .TestLimits.VLo)
   TheExec.Datalog.WriteComment ("Measured Voltage = " & .Read(tlPPMUReadMeasurements))
End If

' Clamp
TheExec.Datalog.WriteComment ("ClampVHi = " & .ClampVHi.Value)
TheExec.Datalog.WriteComment ("ClampVLo = " & .ClampVLo.Value)
If .Gate = tlOn Then
   TheExec.Datalog.WriteComment ("Gate = On")
Else
   TheExec.Datalog.WriteComment ("Gate = Off")
End If
End With

End Function
```

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PPMU Debug Display

PPMU Debug Display

The PPMU Debug Display is an interactive display that shows a block diagram of the pin parametric measurement unit electronics with the programmable parameters in text boxes. Use the display to determine the state of the PPMU hardware and to set PPMU parameters interactively.

This section includes the following topics:

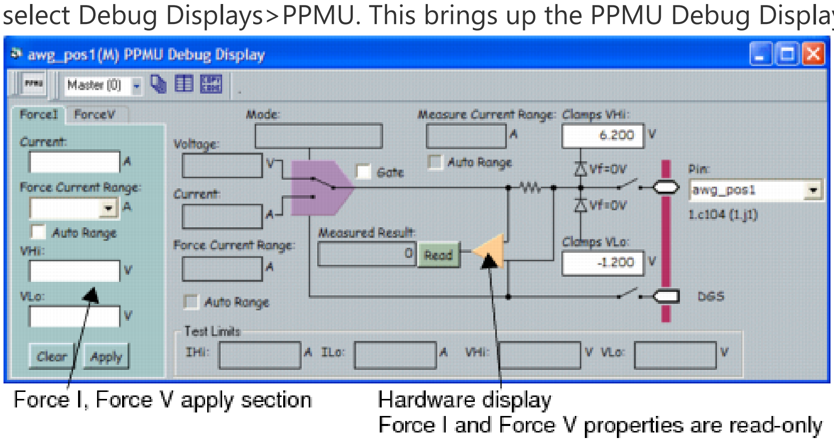
- Starting the Display
- PPMU Debug Display Interface

This section describes features specific to the PPMU Debug Display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

Starting the Display

To launch the PPMU Debug Display, first validate the test program. Then, on the IG-XL Tools tab in the Launch group, select Debug Displays>PPMU. This brings up the PPMU Debug Display as shown in the following figure.



PPMU Debug Display

You can also bring up the PPMU debug display from inside the TDE or using the VBT statement:

TheExec.Tools("PPMU").ShowPin ("<pinname>")

Finally, you can bring up the PPMU debug display from debug displays for instruments that support the PPMU. Click the PPMU box in the following displays:

- AWG Debug Display
- BBAC Capture Debug Display
- BBAC Source Debug Display
- DCIO Channel Debug Display
- HSD1000 Digital Channel Debug Display (HSD1000)
- UltraPin800 Digital Channel Debug Display (UltraPin800)
- UltraPin1600 Digital Channel Debug Display (UltraPin1600)
- Digital Channel Debug Display Interface (UltraPin4000)
- GigaDigHD Board Debug Display
- GigaDig Debug Display
- Starting the Serial Channel Debug Display (SB6G)
- Starting the Serial Channel Debug Display (UltraSerial10G)
- TurboACCapture Debug Display
- TurboACSource Debug Display
- VHFAC Capture Debug Display
- VHFAC Serial Bus Debug Display
- VHFAC Source Debug Display
- VHFAC Time Debug Display

PPMU Debug Display Interface

The PPMU Debug Display is divided into two regions:

- The left side shows ForceI and ForceV properties that you can specify and then apply to the PPMU hardware. Any changes made in the ForceI and ForceV tabs take effect only after you click the Apply button.

- The right side shows the state of the PPMU hardware. Some properties displayed are read-only.

The left side of the display shows the following controls and properties:

Force I	Select this tab to display the Force I, Measure V properties and to modify the properties. For force current values and force current ranges for specific instruments, see PPMU Programming Parameter Values and Ranges. This tab includes the following fields: • Current: Enter the current to force in amps. Programmable current values depend on the instrument. This parameter is required if you select ForceI mode. • Force Current Range: Selectable current ranges depend on the instrument. If you do not specify a range and leave this box blank, the force current range is set to auto range. • Auto Range: If this box is checked, the PPMU selects the range closest to but greater than the programmed current. For instance, if the programmed current is 199A, the 200A range is selected. If 201A is the programmed current, the 2mA range is selected. • VHi, VLo: Voltage test limits. In ForceI mode, these are the limits the voltage must be within to either pass or fail. Specify one limit or both. If you do not specify test limits and leave the box blank, no limits are set and no limit tests are performed.
Force V	Select this tab to display the Force V, Measure I properties and to modify the properties. For force voltage values and measure current ranges for specific instruments, see PPMU Programming Parameter Values and Ranges. This tab includes the following fields: • Voltage: Enter the voltage to force, in volts. Programmable voltage values depend on the instrument. This parameter is required if you select ForceV mode. • Measure Current Range: Selectable measure current ranges depend on the instrument. If you do not specify a range and leave the box blank, the measure current range is set to autorange. • Auto Range: If this box is selected, the Measure Current Range selected by the PPMU is determined from the test limits selected for the particular test. The range chosen is

determined by looking at the absolute value of the upper and lower test limits (IHi and ILo) and selecting a range that is closest to but not less than the limit.

- IHi, ILo: Current test limits. In ForceV mode, these are the limits the current must be within to either pass or fail. You can specify one limit or both. If you do not specify test limits and leave the box empty, no test limits are set and no limit tests are performed.

The top and right side of the display shows the following properties:

Mode	The mode of the PPMU (read-only). • ForceVMeasureI • ForceIMeasureV
Voltage	Displays the programmed voltage (read-only).
Current	Displays the programmed current (read-only).
Force Current Range	Displays the force current range being used (read-only). Available ranges depend on the instrument. See PPMU Programming Parameter Values and Ranges.
Auto Range	This box indicates that you are using the Auto Range selection in the ForceI tab for the force current range and shows the range being used in the field above. Read-only.
Gate	When this box is selected, the PPMU applies the force conditions selected.
Measure Current Range	Displays the measure current range being used (read-only). Available ranges depend on the instrument. See PPMU Programming Parameter Values and Ranges.
Auto Range	This box indicates that you are using the Auto Range selection in the ForceV tab for the measure current range and shows the range being used in the field above. Read-only.
Clamps VHi	High clamp voltage. Voltage clamp ranges depend on the instrument. See PPMU Programming Parameter Values and Ranges. Note that you can select only one set of hardware voltage clamp values, either the digital channel (Vch and Vcl) or PPMU (Clamps VHi and Clamps VLo). If you change Clamps VHi or Clamps VLo values in the PPMU debug display, then IG-XL selects the PPMU clamps, not the digital channel clamps. If you change the Vch or Vcl values in the digital channel debug display, then IG-XL selects the digital channel clamps, not the PPMU clamps.
Clamps VLo	Low clamp voltage.
Measured Result	The last measured result and the result of the test (pass/fail) relative to the test limits selected for the test. Click the Read button to display an updated measurement and test result in this field.
DUT Relay	Connect and disconnect the PPMU DUT (output) relay by selecting the relay on the display.

	Take care when opening and closing PPMU DUT relays using the debug display because IG-XL allows switching under load (hot switching). By default, when the PPMU DUT relay opens or closes, the PPMU DGS ground connection is also opened or closed. For most instruments, disconnecting the PPMU does not affect the gate state. For UltraPin4000 and HPM instruments, however, disconnecting the PPMU also gates off the PPMU.
Pin	The menu allows you to select a pin. Below each pinname is the DIB channel string. Any pin in the pin map that has a PPMU behind it is a valid choice. This channel string is the DIB pin (DibSlotPin) where the PPMU is connected to on the DIB.
Test Limits	The test limits selected for the test. Read-only. If no test limits are programmed, the limits shown are the maximum or minimum programmable current (+1000A, -1000A) and maximum or minimum programmable voltage (+1000V, -1000V). For example, when 1000 is displayed on the HSD PPMU, IG-XL performs the test limits and not the HSD PPMU hardware. On the BBAC PPMU, however, the hardware performs test limits when 1000 is displayed.

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PPMU Programming

PPMU Programming

This section describes programming of a Pin Parametric Measurement Unit (PPMU).

The following topics are available:

- PPMU Reset Values
- PPMU Test Instance Programming
- PPMU VBT Programming

For more information, see:

- PPMU (VBT Syntax Reference)
- PPMU Examples

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PPMU Programming : PPMU Reset Values

PPMU Reset Values

You can reset both the PPMU connections and instrument settings. The table below describes the variations in the PPMU reset states and values among UltraFLEX instruments.

PPMU Reset Values

	Reset Values					
PPMU Setting	AWG6G and GigaDig	BBAC, TurboAC, VHFAC	DCIO	HSD1000/ UltraPin800/ UltraPin1600	HPM/ UltraPin4000/UltraSerial10G	SG6G
Connection State	Disconnected	Disconnected	Connected	Disconnected	Disconnected	Disconnected
Gate	Off	On	On	On	On	On
Mode	FVMI	FVMI	FVMI	FVMI	FVMI	FVMI
Force Current	0.0A	0.0A	0.0A	0.0A	0.0A	0.0A
Force Voltage	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V
Force Current Range	50mA	2mA	32mA	2mA	2mA	2mA
Measure Current Range	50mA	2mA	64mA	2mA	2mA	2mA
Measure Current Test Limits	IHi = 1000A ILo = -1000A	IHi = 1000A ILo = -1000A	IHi = 1000A ILo = -1000A	IHi = 1000A ILo = -1000A	IHi = 1000A ILo = -1000A	IHi = 3.6A ILo = -1.0A

Measure Voltage Test Limits	VHi = 1000V VLo = -1000V	VHi = 1000V VLo = -1000V	VHi = 1000V VLo = -1000V	VHi = 1000V VLo = -1000V	VHi = 1000V VLo = -1000V	VHi = 3.6V VLo = -1.0V
Autoranging Force	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Autoranging Measure	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Voltage Clamp Low	-1.2V (Min)	-3.18V (Min)	-1.3V (Min)	-2.0V (Min) -1.6V (UP1600)	-0.8V (Fixed)[1]	-2.0V (Min)
Voltage Clamp High	6.2V (Max)	8.18V (Max)	6.3V (Max)	6.5V (Max)	1.5V (Fixed)	4.7V (Max)

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1 The UltraPin4000 Voltage Clamps (Low and High) are not programmable.

For information on how to program the reset state, see Reset the PPMU and Reset in the VBT Syntax Reference.

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PPMU Programming : PPMU Test Instance Programming

PPMU Test Instance Programming

You can use the instance editor to create and edit PinPMU_T VBT test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all the test instances for a test program.

To create or edit a PinPMU_T test instance in the instance editor, see Creating and Editing Test Instances. Once you have created a test instance, program the PinPMU_T test instance by filling in the parameter fields. For a reference to the PinPMU_T test parameters, see Pin PMU VBT Test.

For a PPMU programming example using test instances, see PPMU Examples.

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PPMU Programming : PPMU VBT Programming

PPMU VBT Programming

Test programs using Visual Basic for Test (VBT) can have the fastest execution times or minimum test times. The procedure is:

1.	Connect the PPMU.
2.	Set Up PPMU Using Force Current or Force Voltage.
3.	Set the PPMU Voltage Clamps.
4.	Read the Result.
5.	Reset the PPMU.
6.	Disconnect the PPMU.

For information on VBT syntax used to program the PPMU, see PPMU in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Connect the PPMU

Connect the PPMU

Connect the PPMU to the DUT using:

TheHdw.PPMU.Pins("<pinname>").Connect

On the HSD1000, AWG6G, BBAC, GigaDig, TurboAC, and VHFAC instruments, PPMU connections are incremental. That is, connecting the PPMU to the DUT adds the PPMU to any existing connections. It does not disconnect any pre-existing connections, except for DIB Access. DIB Access and PPMU connections are mutually exclusive, which means that connecting the PPMU disconnects DIB Access.

For example, if a DUT pin already has pin electronics connected to the pin, connecting the PPMU to the same DUT pin adds the PPMU connection without disconnecting the pin electronics. If the DUT pin already has DIB Access connections, connecting the PPMU automatically disconnects the DIB Access.

Note that the DCIO, HPM, SB6G, UltraPin800, UltraPin1600, UltraPin4000, and UltraSerial10G instruments do not support PPMU incremental connections.

On the UltraPin800 and UltraPin1600 instrument, you can connect the PPMU and the pin electronics to the DUT simultaneously under certain conditions. For more information, see the usage guidelines listed in the VBT statement, TheHdw.PPMU.AllowPPMUFuncRelayConnection.

See also TheHdw.PPMU.Pins.Connect in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Set Up PPMU Using Force Current or Force Voltage

Set Up PPMU Using Force Current or Force Voltage

Select the mode for the PPMU. The modes can be ForceI or ForceV:

For ForceI:

```
TheHdw.PPMU.Pins("<pinname>").ForceI(Current, ForceCurrentRange,
   TestLimitVHi = 1000, TestLimitVLo = -1000)
```

where:

Current	The value to set for forcing current.
ForceCurrentRange	The current range to set for forcing current.
TestLimitVHi	The high limit to use for a limits test. No limit set if left blank.
TestLimitVLo	The low limit to use for a limits test. No limit set if left blank.

This function sets the ForceCurrentRange and Current to the specified values. It sets the PPMU Mode to tlPPMUForceIMeasureV. If no value is specified for ForceCurrentRange, then IG-XL sets the PPMU to the highest force current range that is supported by the instrument.

For more information, see ForceI in the VBT Syntax Reference.

For ForceV:

```
TheHdw.PPMU.Pins("<pinname>").ForceV(Voltage, MeasureCurrentRange,
   TestLimitIHi = 1000, TestLimitILo = -1000)
```

where:

Voltage	The value to set for forcing voltage.
MeasureCurrentRange	The current range to set for measuring current.
TestLimitIHi	The high limit to use for a limits test. No limit set if left blank.
TestLimitILo	The low limit to use for a limits test. No limit set if left blank.

This function sets the MeasureCurrentRange and Voltage to the specified values. It sets the PPMU Mode to tlPPMUForceVMeasureI. If no value is specified for MeasureCurrentRange, then IG-XL sets the PPMU to the highest measure current range that is supported by the instrument.

For more information, see ForceV in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Set the PPMU Voltage Clamps

Set the PPMU Voltage Clamps

IG-XL allows you to set voltage clamps by programming individual pins:

- TheHdw.PPMU.Pins("").ClampVHi
- TheHdw.PPMU.Pins("").ClampVLo

For more information, see ClampVHi and ClampVLo in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Read the Result

Read the Result

The following syntax causes the PPMU to make a measurement:

Set data1 = TheHdw.PPMU.Pins("<pinname>").Read(tlPPMUReadMeasurements)

This statement returns a PinListData object with doubles.

You can also read back a pass or fail result. If the measured value is between the high and low specified in the test limits, a true is returned. If the measured value is outside the test limits, then a false is returned. The statement is:

Set data1 = TheHdw.PPMU.Pins("<pinname>").Read(tlPPMUReadPassFailResults)

When reading back a pass or fail result, the type that is returned is tlResultType. If the pin passes, the value returned is tlResultPass. If the pin fails, the value returned is tlResultFail.

In the next example, the read function returns one value if all pins are identical. If not, it reads the results from each pin:

```
Dim Pin As IPinData
Dim site As Variant
Set Iface = TheHdw.PPMU.Pins("<pinname>").Read(tlPPMUReadPassFailResults)
  With Iface
    Print #fileWTo, spaceS & "IsUniform: " & .IsUniform
  If .IsUniform Then
    Print #fileWTo, spaceS & "Value: " & .Value
  Else
    For Each Pin In .pins
      If Pin.IsUniform Then
        Print #fileWTo, spaceS & "Pin: " & Pin.Name & vbTab & _
            "Value: " & Pin.Value
    Else
      For Each site In .Sites
        With Pin
          Print #fileWTo, spaceS & "Site: " & site & vbTab & "Pin: " & _
            .Name & vbTab & "Chan: " & _
            .Chan(site) & vbTab & "Value: " & _
            .Value(site)
        End With
      Next site
```

```
        End If
      Next Pin
    End If
    End With
End Sub
```

For more information, see Read in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Reset the PPMU

Reset the PPMU

Restore the PPMU to the default state using the reset VBT statement:

TheHdw.PPMU.Pins("<pinname>").Reset(tlResetConnections)

TheHdw.PPMU.Pins("<pinname>").Reset(tlResetSettings)

The first statement uses the tlResetConnections option to disconnect the PPMU from the DUT. Note that it does not reset the PPMU with default values.

The second statement uses the tlResetSettings option to reset the PPMU with the default values. This statement does not disconnect the PPMU from the DUT.

To both disconnect the PPMU from the DUT and reset the PPMU, use both statements:

TheHdw.PPMU.Pins("<pinname>").Reset(tlResetConnections + tlResetSettings)

For more information, see PPMU Reset Values.

For more information on the VBT syntax, see Reset in the VBT Syntax Reference.

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PPMU Programming : PPMU VBT Programming : Disconnect the PPMU

Disconnect the PPMU

At the end of the test, disconnect the PPMU from the DUT using:

TheHdw.PPMU.Pins("<pinname>").Disconnect

For more information, see Disconnect in the VBT Syntax Reference.

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